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Review

Overview and control strategies of autonomous sailboats—A survey



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ABSTRACT

In recent decades, marine robots have become increasingly important for a wide range of applications in bodies of water, including oceans, gulfs, and lakes. One type of unmanned surface vehicle that has gained attention is the autonomous sailboat, which utilizes wind as a source of propulsion, offering advantages in terms of energy consumption and continuous operation. While autonomous sailboats face limitations in motion and constraints compared to conventional unmanned surface vehicles that use motors, sailing maneuvering techniques can provide motion capability to reach a destination. This survey paper provides a comprehensive overview of autonomous sailboats, including their fundamental background, dynamic models, and guidance systems. Control strategies for heading and speed controls are investigated and categorized based on their purposes and computational criteria. Finally, the potential applications of these control strategies are evaluated and compared, highlighting their importance for future research.

1. Introduction

The ocean covers approximately 70% of the Earth's surface, yet only a small fraction of it has been explored—about 5% or less (Mogias et al., 2021). Ocean exploration poses major challenges, such as environmental disturbances, energy consumption, safety concerns, and the unpredictability of ocean waves. One response to address these challenges has been the development of marine robots. Unmanned Surface Vehicles (USVs)—also known as Autonomous Surface Vehicles (ASVs) or Autonomous Surface Crafts (ASCs)—are marine robots that can operate on the water surface without human intervention. Beyond ocean exploration, USVs have potential applications in energy scavenging, transportation, military operations, and general robotics research (Manley, 2008; Liu et al., 2016). For example, the Saildrone project observed air–sea fluxes and other ocean environments (Gentemann et al., 2020). Another Saildrone project suggested assisting replacing aging buoys that were used to look for indicators of climate-disrupting El Niño events (Voosen, 2018). The Roboat II is designed with a docking capability and inter-connectivity between boats for urban transportation (Wang et al., 2020). In recent years, the military conflict resulting for the Russian invasion of Ukraine has resulted in remote-controlled USVs being used, possibly equipped with weapons (Workman et al., 2022). In addition, USVs have potential in minesweeping operations.

The development of USVs commenced in 1993 when the first USV was built as part of the MIT Sea Grant College Program. This USV, named ARTEMIS, was designed to test the necessary navigation and control systems while collecting simple bathymetric data (Manley, 2008).

USVs can be categorized based on several considerations, such as boat size, hull type, keel type, model type, or control method. Based on the source of propulsion USVs can be categorized into: (1) conventional USVs, whose sources of propulsion are motors or engines; and (2) autonomous sailboats, also known as Autonomous Sailing Vehicles (ASVs), unmanned sailboats, or sailing USVs, whose source of propulsion is wind. Using wind propulsion means that autonomous sailboats have the advantage of less energy consumption than conventional USVs (dos Santos and Goncalves, 2019), as shown in Fig. 1. Another difference between conventional USVs and autonomous sailboats is their simplified models. Since a sailboat needs a sail, which can be either sailcloth or a wingsail, to generate its propulsive force from the wind, this causes substantial roll motion that should be considered. While most conventional USV models are simplified into 3 degrees of freedom (DOF), namely surge, sway, and yaw, most autonomous sailboat models are simplified instead into 4 DOFs, namely surge, sway, roll, and yaw. However, some sailboats, such as a catamaran with a double hull, may neglect the roll motion due to the boat's large base (Sun et al., 2020). Typically, the characteristics of autonomous sailboats can be described as: (1) wind is the only source of propulsion; (2) an autonomous sailboat is not remotely controlled, or the entire control system is onboard; and (3) an autonomous sailboat is entirely energy self-sufficient—this is optional (Stelzer and Jafarmadar, 2011).

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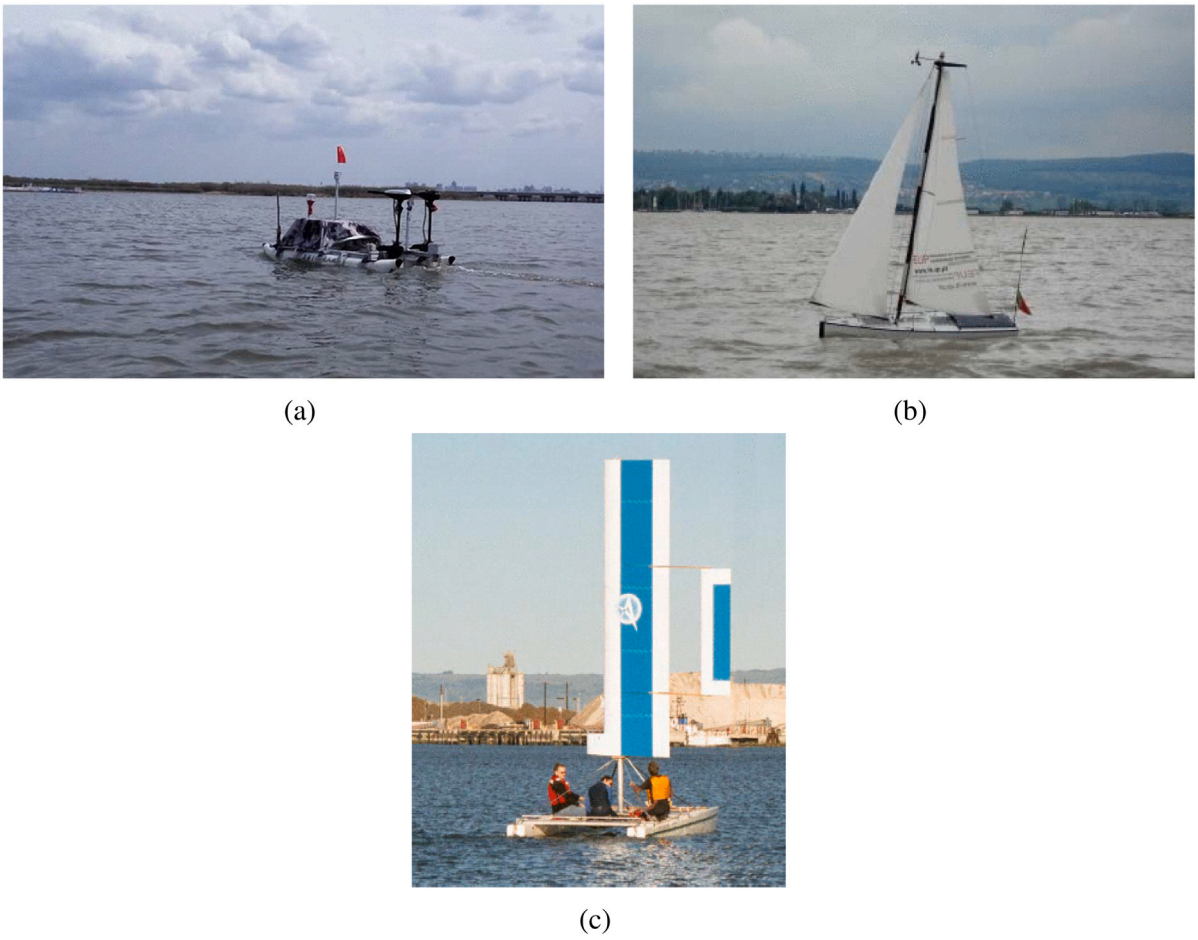


Fig. 1. Conventional USV and autonomous sailboat: (a) Dolphin I—the conventional USV (Liao et al., 2020) (©2020 IEEE); (b) FASt—the autonomous monohull sailboat (Cruz and Alves, 2010) (©2010 IEEE); (c) Atlantis—the autonomous catamaran sailboat (Elkaim and Kelbley, 2006b) (©2006 IEEE).

Additionally, autonomous sailboats can sail for hundreds of days or navigate across the Atlantic Ocean and around Antarctica (An et al., 2021).

An early experiment involving autonomous sailboats was the Atlantis project, involving a catamaran with a wingsail developed by Elkaim in 2001 (Yang et al., 2021; Elkaim, 2021). This sailboat was modified from the Prindle-19 catamaran and guided by GPS. Also, it was used for several kinds of research on autonomous sailboats, such as GPS-based system identification viability testing (Elkaim and Parkinson, 2001), segmented trajectory control and station-keeping (Elkaim and Kelbley, 2006b), and model-free subspace $\mathcal{H}_\infty$ control (Elkaim et al., 2006). Then, the navigational algorithms of Atlantis were applied to develop another hybrid autonomous sailboat named HWT X-1, which was a catamaran with motors and a wingsail modified from the Stiletto catamaran (Lee Boyce and Hugh Elkaim, 2007). Subsequently, other autonomous sailboat works were developed with various models, guidance systems, and control strategies, such as, the FASt project (Cruz and Alves, 2010), the VAIMOS project (Le Bars and Jaulin, 2013), and the OceanVoy project (Sun et al., 2020).

Our team's research was driven by the necessity for pipeline inspection in real-world situations using Autonomous Underwater Vehicles (AUVs) and their sensors (Hoonsuwan et al., 2016). We are engaged in various essential areas of investigation, including: (1) the construction of AUVs on a laboratory scale (Tipsuwan and Hoonsuwan, 2015; Siriyakorn et al., 2019); (2) the development of algorithms that enable the extraction of pipeline information from sonar images through utilization of the self-organizing map (Kasetkasem et al., 2021; Tulsook et al., 2018) and the level set method (Leangaramkul et al., 2019a,b); (3) the creation of algorithms for AUV localization (Kasetkasem et al., 2017); and (4) the implementation of tracking and control mechanisms (Tipsuwan et al., 2019; Rattanawaorahirankul et al., 2017).

The motivation for the current work was our interest in building an autonomous sailboat to ship some loads or spare parts from one place to another using free energy from the wind. Therefore, the main contribution of this paper is to review the control strategies of autonomous sailboats in the past, including the advantages/disadvantages of each control strategy, and the trends of current valuable works. In addition, this paper introduces some terms and motion control techniques related to autonomous sailboats to help interested readers become more familiar with them.

The paper is organized as follows. Section 2 introduces a sailboat's important terms, such as the main components and regions, point of sail, sailing maneuvering techniques, polar diagram, and guidance, navigation, and control (GNC). Then, Section 3 presents sailboat dynamics modeling. The autonomous sailboats' guidance system and path planning are described in Section 4. Motion control strategies for heading and speed controls are presented and described in Section 5. Finally, the conclusions are summarized in Section 6.

2. Overview of sailboats

This section provides an overview of important sailing terminology. First, Section 2.1 covers the terminology related to the parts of a sailboat. Next, Section 2.2 describes the various points of a sail. Following this, Section 2.3 introduces the polar diagram, which is utilized to determine the maximum speed of a sailboat over a true

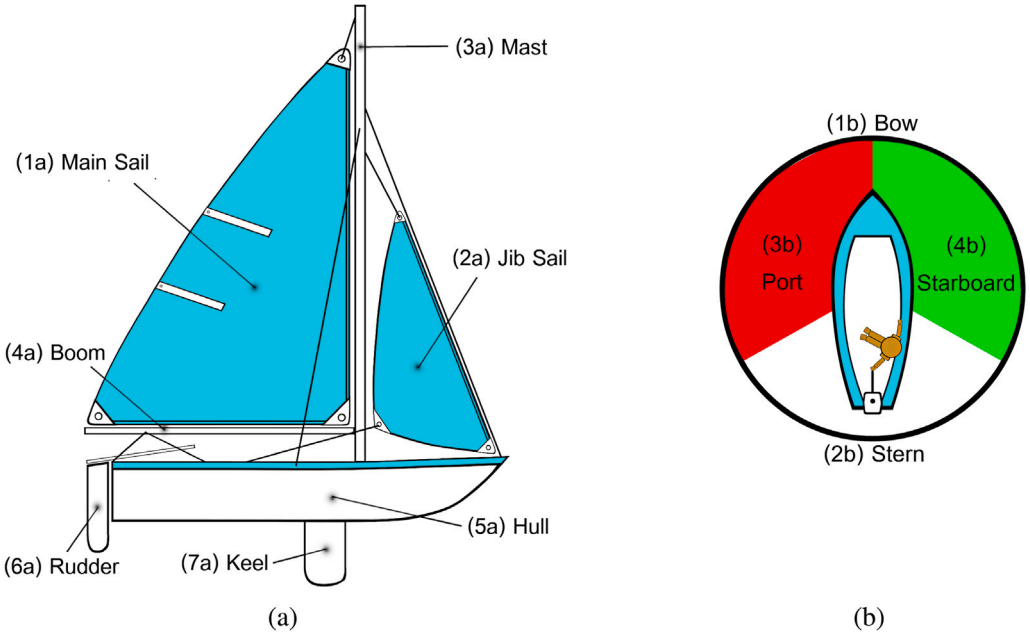


Fig. 2. Parts of a sailboat: (a) components; (b) regions.

Table 1
Parts of a sailboat.

No.	Part	Description
1a	Main Sail	The main part that generates propulsion from the wind. This part can be either sailcloth or a wingsail.
2a	Jib Sail	A smaller part that works in conjunction with the main sail for propulsion when using a sailcloth.
3a	Mast	The vertical pole that supports the sail.
4a	Boom	The horizontal rod that supports the sail.
5a	Hull	The main body of the boat.
6a	Rudder	The part that changes the heading direction.
7a	Keel	The part that prevents motions caused by the side force.
1b	Bow	The front part of the boat.
2b	Stern	The back part of the boat.
3b	Port	The left side of the boat.
4b	Starboard	The right side of the boat.

wind angle. Then, Section 2.4 explains sailing maneuvering techniques for upwind and downwind directions. Lastly, Section 2.5 covers the three subsystems involved in the movement of vehicles, particularly autonomous sailboats, which are guidance, navigation, and control.

2.1. Parts of a sailboat

Different types of sailboats consist of various components. The main components are the sail, mast, boom, hull, rudder, and keel, while the regions of a sailboat are the bow, stern, port, and starboard. These parts are commonly applied in a monohull sailboat, are shown in Fig. 2 and explained in Table 1.

2.2. Point of sail

Since the main source of a sailboat's propulsion is wind, possible sailing directions are limited. In sailing theory, the point of sail or the wind circle is the heading direction of a sailboat with respect to the incoming direction of the true wind, as shown in Fig. 3. The point of sail describes the ability to sail a sailboat in each situation.

Primarily, a sailboat cannot sail directly into the wind, or it cannot sail while in the no-go zone, where there is no propulsive force generated. If the destination is too close to the direction of the incoming

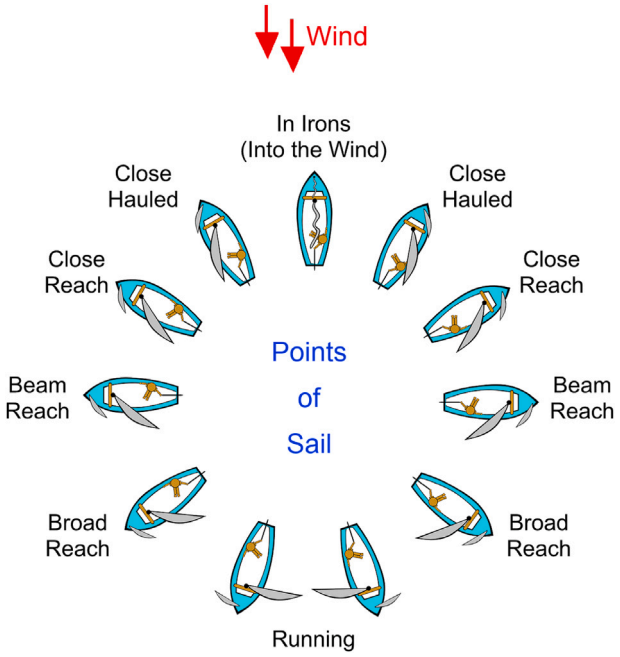


Fig. 3. Points of sail.

wind or a sailboat is sailing upwind, a sailing maneuvering technique called tacking must be considered. On the other hand, if a sailboat is running downwind, the sail acts as a parachute to generate the drag force, and the sailboat's speed will be limited, not to exceed the wind speed because the apparent wind speed reduces as the sailboat speed increases downwind. To exceed this limited speed, a sailing maneuvering technique called jibing must be taken into account (Stelzer et al., 2007; Rousmaniere, 2014; Kimball, 2009). Section 2.4 provides details of upwind and downwind maneuvering techniques.

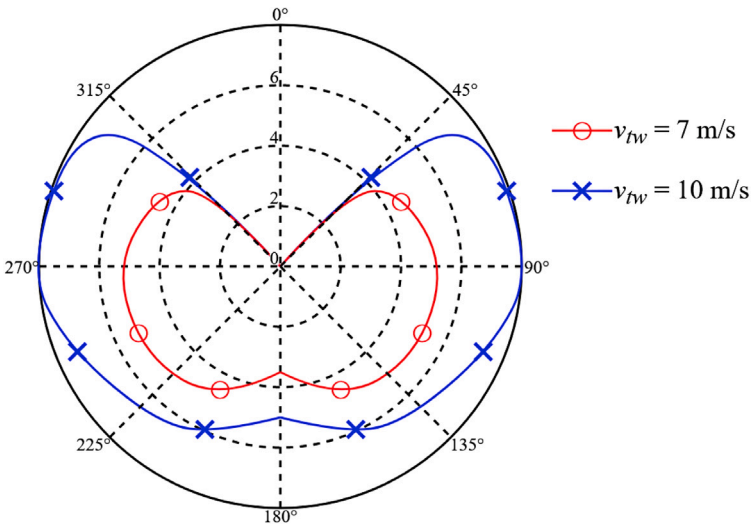


Fig. 4. Polar diagram of a sailboat at true wind speed v_{tw} of 7 m/s and 10 m/s.

2.3. Polar diagram

A polar diagram is a graph used to determine the maximum speed of a sailboat over true wind angles and speeds related to the sailboat heading (Stelzer and Pröll, 2008). This graph can be obtained by either experimentation or by solving the sailboat model. A solving method for a polar diagram called Quantified Set Inversion (QSI) has been proposed (Herrero et al., 2010). This polar diagram was applied in many sailboat papers to determine sailboat potential or for use as a lookup table for speed control (Briere, 2008; Plumet et al., 2015; Pêtrès et al., 2011; Elkaim and Kelbley, 2006a; Abrougui et al., 2019b; Saoud et al., 2015b; Treichel and Jouffroy, 2010; Xiao et al., 2012; Shen et al., 2019).

A polar diagram is also helpful in determining a sailboat's maximum Velocity-Made-Good (VMG). The VMG is the sailboat's speed in the parallel direction of the wind, either upwind or downwind. Notably, the VMG is different from the Velocity-Made-good-on-Course (VMC), which is the sailboat speed in the direction of the target (Jing et al., 2020).

Fig. 4 shows an example of a polar diagram. Notably, the sailboat surge speed tends towards zero when sailing towards the wind in the no-go zone (as described in Section 2.2), as well as plots in the polar diagram. Some literature may describe or use a polar diagram slightly differently, with a constant true wind speed and plots of sail angles (Saoud et al., 2014).

Therefore, the next section introduces two sailing techniques for maneuvering in upwind and downwind directions.

2.4. Sailing maneuvering techniques for upwind and downwind directions

As outlined in Section 2.2, sailboats encounter challenges while sailing upwind and downwind. In upwind sailing, direct sailing against the wind is infeasible, while in downwind sailing, the sailboat's speed decreases. To address these concerns, specific sailing maneuvers are applied.

There are two maneuvering techniques for sailing in upwind and downwind directions. Upwind sailing refers to the scenario where a sailboat navigates against the wind, with the wind direction blowing around the bow. In contrast, downwind sailing describes the scenario where a sailboat travels in the same direction as the wind, with the wind direction blowing around the stern.

Tacking is a sailing maneuvering technique to sail towards the wind (upwind) by turning the bow towards the wind from the port to the starboard or vice versa. The sailboat will steer diagonally in the wind direction. Performing multiple tacks can reach the destination in the incoming wind direction based on a zigzag course, as shown in Fig. 5(a).

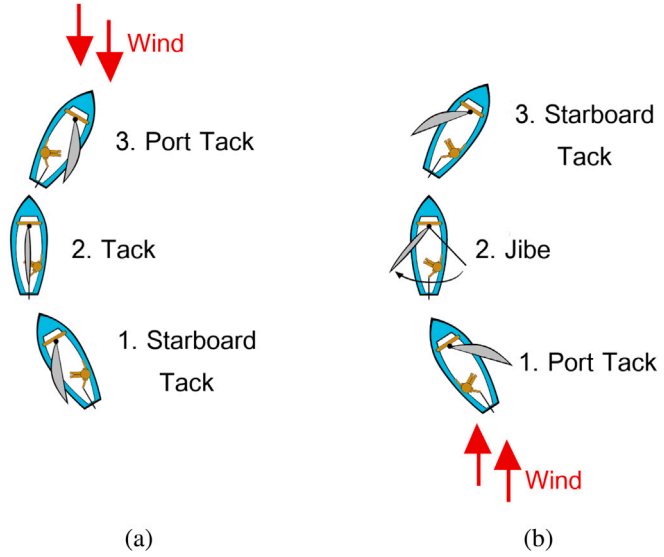


Fig. 5. Sailing maneuver techniques: (a) tacking from starboard tack to port tack; (b) jibing from port tack to starboard tack.

Similarly, jibing (or gybing) is a sailing maneuvering technique like tacking, but instead, the sailboat is moving in the same direction as the wind (downwind) that turns the stern instead (Stelzer et al., 2007; Kimball, 2009), as shown in Fig. 5(b).

2.5. Guidance, Navigation and Control

Guidance, Navigation, and Control (GNC), as shown in Fig. 6, are interrelated subsystems that work together to control the movement of vehicles, such as ships, aircraft, or autonomous robots, and particularly autonomous sailboats (Liu et al., 2016; Fossen, 2011).

(1) The guidance subsystem involves a mission planner and global/local path planner, including the vehicle's trajectory to reach its destination while also considering various constraints such as: obstacles, vehicle dynamics, and environmental factors. In case of unexpected events, the guidance system must be able to re-plan and update the path

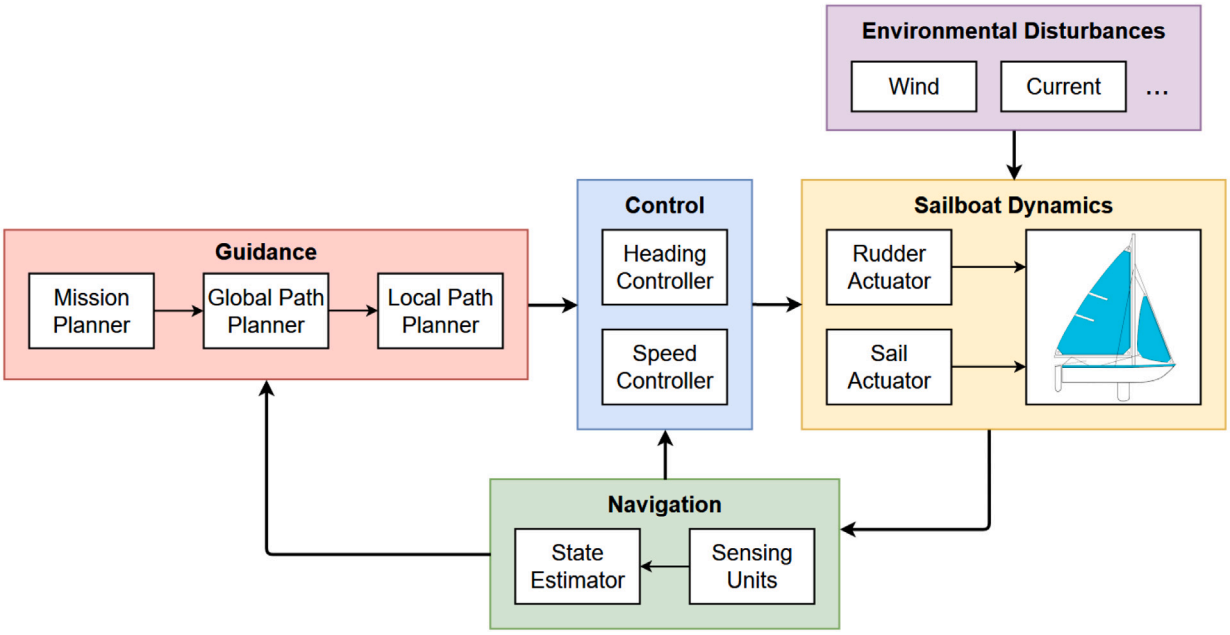


Fig. 6. GNC system of autonomous sailboat.

with the desired trajectory in real-time. Then, the guidance subsystem sends the desired trajectory to the control subsystem.

(2) The navigation subsystem gathers information from various sensors and uses the information to identify vehicle states (position, velocity, and orientation) using an estimation algorithm. Then, the navigation subsystem sends the estimated states to the guidance and control subsystems. The navigation subsystem enables the vehicle to accurately determine its location and estimate the distance to its destination.

(3) The control subsystem determines a proper control signal to the vehicle's actuators, such as motors, thrusters, or rudders, using the designed trajectory from the guidance system and the estimated states, including sensor information from the navigation subsystems. After that, the control signal is sent to the vehicle actuators to generate forces and moments to move the vehicle to achieve the desired trajectory.

These three subsystems cooperate and communicate together to achieve the stated mission of the sailboat. As stated earlier, the guidance subsystem generates the reference trajectory based on the mission and information received from the navigation subsystem. Then, the reference trajectory is sent to the control subsystem. Finally, the control subsystem determines the proper control signal and sends the control signal to the sailboat actuators to move the sailboat to track the reference trajectory.

3. Dynamic model of sailboats

There are two well-known sailboat dynamic models: Jaulin's model with 3 DOF and some sailboat coefficients that exclude the roll motion (Jaulin, 2004), and Xiao's model with 4 DOF (Xiao and Jouffroy, 2014). Xiao's model is derived from Fossen's model (Fossen, 2011, 1994, 2002), which describes the kinematics and dynamics of general marine craft and has been adapted and used in numerous works for USVs, sailboats, and other marine craft. Some examples of these works are (dos Santos and Goncalves, 2019; Saoud et al., 2014, 2015a; Wille et al., 2016a). In this paper, Xiao's model is presented as it is more comprehensive and has more DOFs.

The notation system used in the dynamic model of sailboats is presented in Table 2. These notations are based on SNAME (SNAME, 1950), as well as other notations described in Fossen (2011) and Xiao and Jouffroy (2014). To provide a clear understanding of sailboat movement, the North-East-Down (NED) coordinate system $(x_n y_n z_n)$ is

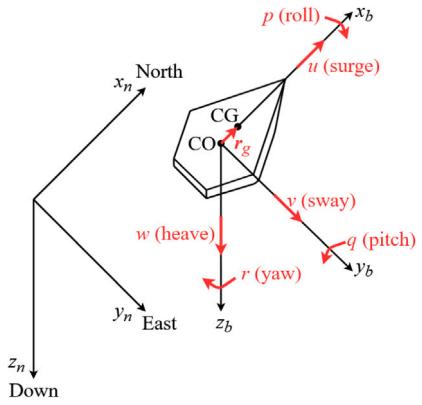


Fig. 7. Sailboat coordinate systems.

initially chosen to define the reference frame (n-frame). This n-frame is an earth-fixed frame. Subsequently, the moving coordinate system $(x_b y_b z_b)$ of the body-fixed frame (b-frame) is fixed to the sailboat body. For clarity, the n-frame and the b-frame are illustrated in Fig. 7. Additionally, Figs. 8 and 9 depict the sailboat movement and wind direction, respectively.

Two dynamic models are introduced in this paper: the full dynamic model and the simplified dynamic model. The next subsection presents the full dynamic model of a sailboat, while the one after that introduces the simplified dynamic model.

3.1. Full dynamic model

A dynamic model of a sailboat consists of two equations of motion: a kinematic equation, which describes only the geometrical aspects of motion between the n-frame and the b-frame, and a kinetic equation, which describes the relationship between forces and motion. The equations of motion of a sailboat can be expressed using:

$$\dot{\boldsymbol{\eta}} = \boldsymbol{J}(\boldsymbol{\eta})\boldsymbol{\nu} \tag{1}$$

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} + \boldsymbol{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \boldsymbol{D}(\boldsymbol{\nu},\boldsymbol{\eta}) + \boldsymbol{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} \tag{2}$$

Table 2
Notation for sailboat dynamic model.

Notation	Description
A	Plan area of the foil
$\boldsymbol{C}, \boldsymbol{C}_{RB}, \boldsymbol{C}_A$	System/rigid body/added mass Coriolis-centripetal matrix
C_D	Drag coefficient
C_{Di}	Induced drag coefficient
C_L	Lift coefficient
CG	Center of gravity of the sailboat
CO	Origin of the b-frame
CoE	Center of effort of the foil
$\boldsymbol{D}, \boldsymbol{D}_k, \boldsymbol{D}_h, \boldsymbol{D}_{heel}, \boldsymbol{D}_{yaw}$	Vector of system/keel/hull/heel/yaw damping
D	Drag force acting on the foil
D_k, D_s, D_r	Drag force acting on the keel/sail/rudder
F_{rh}	Hull resistance
$\boldsymbol{g}$	Vector of restoring forces
I	Moment of inertia of the sailboat in the b-frame
$\boldsymbol{J}$	Coordinate transformation matrix
L	Lift forces acting on the foil
L_k, L_s, L_r	Lift force acting on the keel/sail/rudder
$\boldsymbol{M}, \boldsymbol{M}_{RB}, \boldsymbol{M}_A$	System/rigid body/added mass inertia matrix
m	Total mass of the sailboat
p, q, r	Angular velocity of the sailboat about x, y, z-axis (roll, pitch, yaw) in the b-frame
$\boldsymbol{r}_g$	Vector of CG position in the b-frame
s	Span of the foil
u, v, w	Linear velocity of the sailboat in x, y, z-direction (surge, sway, heave) in the b-frame
$\boldsymbol{v}$	Vector of linear velocity in the b-frame
v_a	Apparent speed in the b-frame
$v_{ak}, v_{ah}, v_{aw}, v_{ar}$	Apparent speed of the keel/hull/wind/rudder in the b-frame
v_{a*u}, v_{a*v}	Apparent surge, sway speed of the * (keel k, hull h, wind w, rudder r) in the b-frame
v_{tw}	True wind speed in the n-frame
$X/Y/K/N_{\dot{u}-\dot{r}}$	Added mass coefficients in the b-frame
x, y, z	Position of the sailboat in x, y, z-axis in the n-frame
x_*, y_*, z_*	CoE position of the $*$ (keel k, hull h, sail s, rudder r) in the b-frame
$x_b y_b z_b$	Body-fixed reference frame (b-frame)
x_m	X-coordinate of the mast in the b-frame
$x_n y_n z_n$	Inertial reference frame (n-frame)
x_{sm}	Distance between the mast and the sail's CoE
α	Angle of attack of the foil in the b-frame
$\alpha_{ak}, \alpha_{ah}, \alpha_{aw}, \alpha_{ar}$	Apparent angle of the keel/hull/wind/rudder in the b-frame
$\alpha_k, \alpha_s, \alpha_r$	Angle of attack of the keel/sail/rudder in the b-frame
α_{tw}	True wind angle in the n-frame
β	Sideslip angle in the n-frame
χ	Course angle in the n-frame
δ_r	Rudder angle in the b-frame
δ_s	Sail angle in the b-frame
$\boldsymbol{\eta}$	Vector of position and orientation of the sailboat in the n-frame
$\boldsymbol{\nu}$	Vector of linear and angular velocities of the sailboat in the b-frame
$\boldsymbol{\omega}$	Vector of angular velocity of the sailboat in the b-frame
ϕ, θ, ψ	Euler's angle of the sailboat about x, y, z-axis (roll, pitch, yaw) in the n-frame
ρ	Fluid density
$\boldsymbol{\tau}, \boldsymbol{\tau}_s, \boldsymbol{\tau}_r$	Vector of system/sail/rudder propulsive forces

where $\boldsymbol{\eta}$ is the $n \times 1$ position/orientation vector, $\boldsymbol{\nu}$ is the $n \times 1$ linear/angular velocity vector, $\boldsymbol{J}(\boldsymbol{\eta})$ is the transformation matrix, $\boldsymbol{\tau}$ is the $n \times 1$ propulsive force/moment vector, $\boldsymbol{M}$ is the $n \times n$ system inertia matrix, $\boldsymbol{C}(\boldsymbol{\nu})$ is the $n \times n$ system Coriolis-centripetal matrix, $\boldsymbol{D}(\boldsymbol{\nu}, \boldsymbol{\eta})$ is $n \times 1$ the system damping force vector, and $\boldsymbol{g}(\boldsymbol{\eta})$ is the $n \times 1$ gravitational/restoring force vector.

Notably, n is equal to 6 for a full dynamic model with $\boldsymbol{\eta} = [x, y, z, \phi, \theta, \psi]^\top$ and $\boldsymbol{\nu} = [u, v, w, p, q, r]$. However, under the assumptions given in the next subsection, the full dynamic model can be simplified by omitting the heave and pitch motions. Thus, n is reduced to 4 for the simplified dynamic model with $\boldsymbol{\eta} = [x, y, \phi, \psi]^\top$ and $\boldsymbol{\nu} = [u, v, p, r]^\top$.

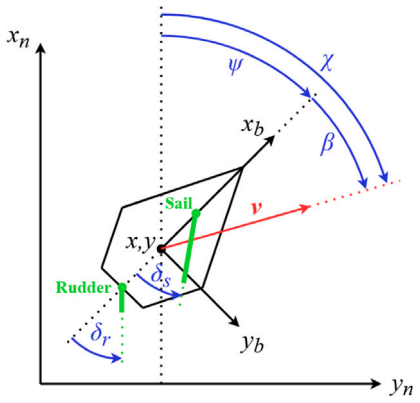


Fig. 8. Sailboat movement vectors.

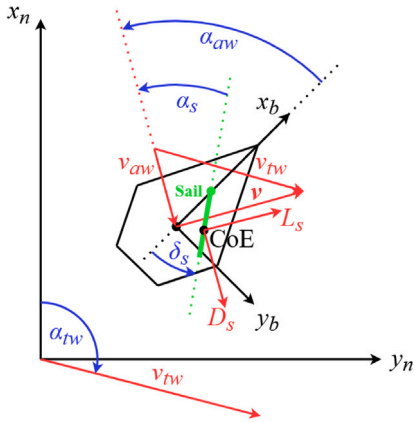


Fig. 9. Wind vectors acting on a sailboat.

3.2. Simplified dynamic model

Sailboats are capable of moving in 6 DOFs with both nonlinear and complex behavior. However, through the application of maneuvering theory and specific assumptions, these movements can be simplified into a 4-DOF planar motion (or 3-DOF for catamarans). In this paper, we introduce a simplified 4-DOF dynamic model based on the following assumptions (Fossen, 2011; Xiao and Jouffroy, 2014):

1. The sailboat is treated as a rigid body. Consequently, the rate of change in the distance between the origin of the b-frame (CO) and the Center of Gravity (CG) of the sailboat is zero ($\dot{\boldsymbol{r}}_g = \boldsymbol{0}$). In addition, the CO is defined at the same position as the CG ($\boldsymbol{r}_g = \boldsymbol{0}$).
2. The heave and pitch motions, w and q, respectively, are considered to be small and are approximated as zero ($w = q = 0$). By omitting the heave and pitch motions, the sailboat motion is simplified into a 4-DOF model, enabling the sailboat motion to be analyzed only in the surge, sway, roll, and yaw directions. (For catamarans, the roll direction can be ignored.)
3. The water is assumed to be calm, with the water current assumed to be zero. In addition, the sailboat motions influenced by waves and environmental disturbances are ignored.
4. Added mass coefficients (hydrodynamic derivatives) are treated as constants.
5. The resultant force directly applies at a single point of the foil, the aerodynamic Center of Effort (CoE), which is assumed to remain constant for each foil.
6. The sailboat is symmetrical about the $x_b z_b$-plane, which results in $\boldsymbol{M}_{A_{ij}} = \boldsymbol{M}^T_{A_{ij}}$.

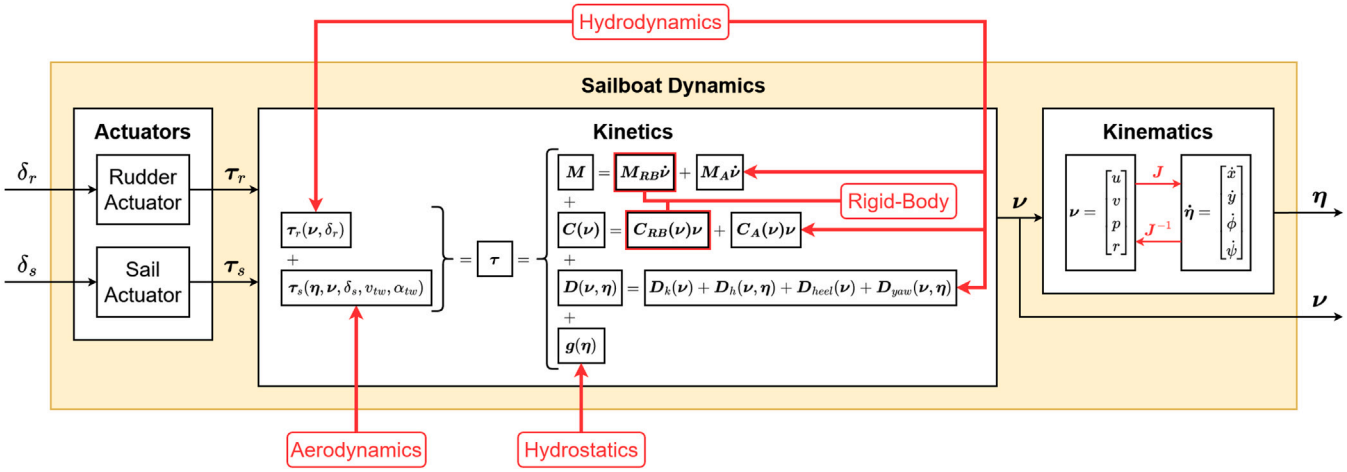


Fig. 10. Simplified sailboat dynamic model diagram.

Based on these assumptions, the simplified model can be written as:

$$\dot{\boldsymbol{\eta}} = \boldsymbol{J}(\boldsymbol{\eta})\boldsymbol{\nu} \tag{3}$$

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} + \boldsymbol{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \boldsymbol{D}(\boldsymbol{\nu},\boldsymbol{\eta}) + \boldsymbol{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} \tag{4}$$

where $\boldsymbol{\eta} = [x, y, \phi, \psi]^\top$ is the position/orientation vector, $\boldsymbol{\nu} = [u, v, p, r]^\top$ is the linear/angular velocity vector, and $\boldsymbol{\tau}$ is the propulsive force/moment acting on the sailboat, which can be controlled by the sail angle δ_s and the rudder angle δ_r as shown in Fig. 10. The expression of $\boldsymbol{\tau}$ is described later in Eq. (26).

The transformation matrix $\boldsymbol{J}(\boldsymbol{\eta})$ is given by:

$$\boldsymbol{J}(\boldsymbol{\eta}) = \begin{bmatrix} \cos(\psi) & -\sin(\psi)\cos(\phi) & 0 & 0 \\ \sin(\psi) & \cos(\psi)\cos(\phi) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \cos(\phi) \end{bmatrix}. \tag{5}$$

The system inertia matrix $\boldsymbol{M}$ and the system Coriolis-centripetal matrix $\boldsymbol{C}(\boldsymbol{\nu})$ include the added mass effects that cause hydrodynamic forces and moments to act as an additional virtual mass to the rigid body due to the acceleration of the boat and the inertia of the surrounding fluid. Thus, $\boldsymbol{M}$ and $\boldsymbol{C}(\boldsymbol{\nu})$ are given by:

$$\boldsymbol{M} = \boldsymbol{M_{RB}} + \boldsymbol{M_A} \tag{6}$$

$$\boldsymbol{C}(\boldsymbol{\nu}) = \boldsymbol{C_{RB}}(\boldsymbol{\nu}) + \boldsymbol{C_A}(\boldsymbol{\nu}) \tag{7}$$

where $\boldsymbol{M_{RB}}$ is the rigid body inertia matrix, $\boldsymbol{M_A}$ is the added mass inertia, $\boldsymbol{C_{RB}}(\boldsymbol{\nu})$ is the rigid body Coriolis-centripetal matrix, and $\boldsymbol{C_A}(\boldsymbol{\nu})$ is the added mass Coriolis-centripetal matrix. All four matrices can be defined as:

$$\boldsymbol{M_{RB}} = \begin{bmatrix} m & 0 & 0 & 0 \\ 0 & m & 0 & 0 \\ 0 & 0 & I_{xx} & 0 \\ 0 & 0 & 0 & I_{zz} \end{bmatrix} \tag{8}$$

$$\boldsymbol{M_A} = -\begin{bmatrix} X_{\dot{u}} & 0 & 0 & 0 \\ 0 & Y_{\dot{v}} & 0 & 0 \\ 0 & 0 & K_{\dot{p}} & 0 \\ 0 & 0 & 0 & N_{\dot{r}} \end{bmatrix} \tag{9}$$

$$\boldsymbol{C_{RB}}(\boldsymbol{\nu}) = \begin{bmatrix} 0 & -mr & 0 & 0 \\ mr & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{10}$$

$$\boldsymbol{C_A}(\boldsymbol{\nu}) = \begin{bmatrix} 0 & 0 & 0 & Y_{\dot{v}}v \\ 0 & 0 & 0 & -X_{\dot{u}}u \\ 0 & 0 & 0 & 0 \\ -Y_{\dot{v}}v & X_{\dot{u}}u & 0 & 0 \end{bmatrix}. \tag{11}$$

Notably, the added mass inertia matrix $\boldsymbol{M_A}$ is simplified by neglecting the off-diagonal elements ($Y_{\dot{p}} = Y_{\dot{r}} = K_{\dot{r}} = 0$).

Before considering the system damping forces $\boldsymbol{D}(\boldsymbol{\nu},\boldsymbol{\eta})$, two general formulas are introduced for the lift force L and the drag force D that are given by:

$$L = \frac{1}{2}\rho A v_a^2 C_L(\alpha) \tag{12}$$

$$D = \frac{1}{2}\rho A v_a^2 \left(C_D(\alpha) + C_{Di}(\alpha)\right) \tag{13}$$

where ρ is the fluid density, A is the plan area of the foil, v_a is the apparent speed, $C_L(\alpha)$ is the lift coefficient, $C_D(\alpha)$ is the drag coefficient, $C_{Di}(\alpha)$ is the induced drag coefficient, and α is the angle of attack between the apparent incoming fluid and the foil. Notably, $C_L(\alpha)$, $C_D(\alpha)$, and $C_{Di}(\alpha)$ are functions of α, while A remains constant for the foil. Both $C_L(\alpha)$ and $C_D(\alpha)$ can be obtained from wind tunnel experiments or approximated using even sine or cosine functions (Melin, 2015; Apostolyuk, 2007). The induced drag coefficient $C_{Di}(\alpha)$ is given by:

$$C_{Di}(\alpha) = \frac{C_L^2(\alpha)A}{\pi s^2} \tag{14}$$

where s is the span of the foil.

The following letters/words will be used as forces generated by that component or movement of the sailboat: k represents the keel, h represents the hull, *heel* represents the movement in the heel (row) direction, and *yaw* represents the movement in the yaw direction. For example, L_k is the lift force generated by the keel.

Therefore, the system damping forces $\boldsymbol{D}(\boldsymbol{\nu},\boldsymbol{\eta})$ due to the sailboat velocities and orientations can be found using:

$$\boldsymbol{D}(\boldsymbol{\nu},\boldsymbol{\eta}) = \boldsymbol{D}_k(\boldsymbol{\nu}) + \boldsymbol{D}_h(\boldsymbol{\nu},\boldsymbol{\eta}) + \boldsymbol{D}_{heel}(\boldsymbol{\nu}) + \boldsymbol{D}_{yaw}(\boldsymbol{\nu},\boldsymbol{\eta}) \tag{15}$$

where $\boldsymbol{D}_k(\boldsymbol{\nu})$ and $\boldsymbol{D}_h(\boldsymbol{\nu},\boldsymbol{\eta})$ are the damping forces due to the keel and the hull of the sailboat, respectively. Furthermore, $\boldsymbol{D}_{heel}(\boldsymbol{\nu})$ and $\boldsymbol{D}_{yaw}(\boldsymbol{\nu},\boldsymbol{\eta})$ are the damping forces due to the movements in the heel and the yaw directions of the sailboat, respectively. All of the damping forces are given by:

$$\boldsymbol{D}_k(\boldsymbol{\nu}) = \begin{bmatrix} -L_k\sin(\alpha_{ak}) + D_k\cos(\alpha_{ak}) \\ -L_k\cos(\alpha_{ak}) - D_k\sin(\alpha_{ak}) \\ \left(-L_k\cos(\alpha_{ak}) - D_k\sin(\alpha_{ak})\right)|z_k| \\ \left(L_k\cos(\alpha_{ak}) + D_k\sin(\alpha_{ak})\right)|x_k| \end{bmatrix} \tag{16}$$

$$\boldsymbol{D}_h(\boldsymbol{\nu},\boldsymbol{\eta}) = \begin{bmatrix} F_{rh}(v_{ah})\cos(\alpha_{ah}) \\ -F_{rh}(v_{ah})\sin(\alpha_{ah})\cos(\phi) \\ -F_{rh}(v_{ah})\sin(\alpha_{ah})\cos(\phi)|z_h| \\ F_{rh}(v_{ah})\sin(\alpha_{ah})\cos(\phi)|x_h| \end{bmatrix} \tag{17}$$

$$\boldsymbol{D}_{heel}(\boldsymbol{\nu}) + \boldsymbol{D}_{yaw}(\boldsymbol{\nu}, \boldsymbol{\eta}) = \begin{bmatrix} 0 \\ 0 \\ a\dot{\phi}|\dot{\phi}| \\ b\dot{\phi}|\dot{\phi}|\cos(\phi), \end{bmatrix} \tag{18}$$

where α_{ak} is the keel's apparent angle, z_k is the z-coordinate of the keel's CoE position in the b-frame, x_k is the x-coordinate of the keel's CoE position in the b-frame, F_{rh} is the hull resistance as a function of the hull's apparent speed v_{ah}, α_{ah} is the hull's apparent angle, z_h is the z-coordinate of the hull's CoE position in the b-frame, and x_h is the x-coordinate of the hull's CoE position in the b-frame. Furthermore, a and b are constants obtained using the inclining test and fitting a quadratic (Philpott et al., 1993). The keel's angle of attack α_k, the keel's apparent angle α_{ak}, and the keel's apparent speed v_{ak} are determined using:

$$\alpha_k = \alpha_{ak} = \arctan2(v_{akv}, -v_{aku}) \tag{19}$$

$$v_{ak} = \sqrt{v_{aku}^2 + v_{akv}^2} \tag{20}$$

where $v_{aku} = -u + ry_k$ and $v_{akv} = -v + rx_k + pz_k$. In addition, the hull's apparent angle α_{ah} and the hull's apparent speed v_{ah} are determined using:

$$\alpha_{ah} = \arctan2(v_{ahv}, -v_{ahu}) \tag{21}$$

$$v_{ah} = \sqrt{v_{ahu}^2 + v_{ahv}^2} \tag{22}$$

where $v_{ahu} = -u + ry_h$ and $v_{ahv} = (-v - rx_h + pz_h)\sec(\phi)$.

The gravitational/restoring force vector $\boldsymbol{g}(\boldsymbol{\eta})$ is given by:

$$\boldsymbol{g}(\boldsymbol{\eta}) = \begin{bmatrix} 0 \\ 0 \\ c\phi^2 + d\phi \\ 0 \end{bmatrix} \tag{23}$$

where c and d are constants obtained in the same way as a and b.

Now, consider the propulsive force/moment $\boldsymbol{\tau}_s$, which is generated by the wind acting on the sail. Let v_{tw} be the true wind speed in the n-frame with the direction α_{tw}. Thus, the apparent wind angle α_{aw} and the apparent wind speed v_{aw} are given by:

$$\alpha_{aw} = \arctan2(v_{awv}, -v_{awu}) \tag{24}$$

$$v_{aw} = \sqrt{v_{awu}^2 + v_{awv}^2} \tag{25}$$

where $v_{awu} = v_{tw}\cos(\alpha_{tw} - \psi) - u + ry_s$ and $v_{awv} = v_{tw}\sin(\alpha_{tw} - \psi)\cos(\phi) - v - rx_s + pz_s$.

The propulsive force/moment $\boldsymbol{\tau}$ acting on the sailboat is given by:

$$\begin{aligned} \boldsymbol{\tau} &= \boldsymbol{\tau}(\boldsymbol{\eta}, \boldsymbol{\nu}, \delta_r, \delta_s, v_{tw}, \alpha_{tw}) \\ &= \boldsymbol{\tau}_s(\boldsymbol{\eta}, \boldsymbol{\nu}, \delta_s, v_{tw}, \alpha_{tw}) + \boldsymbol{\tau}_r(\boldsymbol{\nu}, \delta_r) \end{aligned} \tag{26}$$

where $\boldsymbol{\tau}_s(\boldsymbol{\eta}, \boldsymbol{\nu}, \delta_s, v_{tw}, \alpha_{tw})$ and $\boldsymbol{\tau}_r(\boldsymbol{\nu}, \delta_r)$ are the forces/moments generated by the sail and rudder, respectively, which can be found using:

$$\boldsymbol{\tau}_s(\boldsymbol{\eta}, \boldsymbol{\nu}, \delta_s, v_{tw}, \alpha_{tw}) = \begin{bmatrix} L_s\sin(\alpha_{aw}) - D_s\cos(\alpha_{aw}) \\ L_s\cos(\alpha_{aw}) + D_s\sin(\alpha_{aw}) \\ \big(L_s\cos(\alpha_{aw}) + D_s\sin(\alpha_{aw})\big)|z_s| \\ -\big(L_s\sin(\alpha_{aw}) - D_s\cos(\alpha_{aw})\big)x_{sm}\sin(\delta_s) + \cdots \\ \big(L_s\cos(\alpha_{aw}) + D_s\sin(\alpha_{aw})\big)\big(x_m - x_{sm}\cos(\delta_s)\big) \end{bmatrix} \tag{27}$$

$$\boldsymbol{\tau}_r(\boldsymbol{\nu}, \delta_r) = \begin{bmatrix} L_r\sin(\alpha_{ar}) - D_r\cos(\alpha_{ar}) \\ L_r\cos(\alpha_{ar}) + D_r\sin(\alpha_{ar}) \\ \big(L_r\cos(\alpha_{ar}) + D_r\sin(\alpha_{ar})\big)|z_r| \\ -\big(L_r\cos(\alpha_{ar}) + D_r\sin(\alpha_{ar})\big)|x_r| \end{bmatrix} \tag{28}$$

where z_s is the z-coordinate of the sail's CoE position, x_{sm} is the distance between the mast and the sail's CoE, δ_s is the sail angle, x_m is the x-coordinate of the mast, α_{ar} is the rudder's apparent angle, z_r is the z-coordinate of the rudder's CoE position, and x_r is the x-coordinate of the rudder's CoE position. The sail's angle of attack α_s can be determined using:

$$\alpha_s = \alpha_{aw} - \delta_s. \tag{29}$$

In addition, the rudder's apparent angle α_{ar}, the rudder's angle of attack α_r, and the rudder's apparent speed v_{ar} can be determined using:

$$\alpha_{ar} = \arctan2(v_{arv}, -v_{aru}) \tag{30}$$

$$\alpha_r = \alpha_{ar} - \delta_r \tag{31}$$

$$v_{ar} = \sqrt{v_{aru}^2 + v_{arv}^2} \tag{32}$$

where $v_{aru} = -u + ry_r$ and $v_{arv} = -v - rx_r + pz_r$.

In summary, the sailboat uses the wind as an energy source for the propulsive force/moment $\boldsymbol{\tau}$, which can be controlled by the sail angle δ_s and the ruddle angle δ_r as shown in Fig. 10. The position/orientation $\boldsymbol{\eta}$ and the linear/angular velocity $\boldsymbol{\nu}$ are the outputs of the simplified sailboat dynamic model.

In preparation for the discussion on motion control strategies in Section 5, the following section introduces a guidance system. One of the primary functions of a guidance system is to generate desired values, such as the desired heading ψ_d and speed u_d, which are subsequently sent to the heading and speed controllers, as illustrated in Fig. 11.

4. Guidance system

A guidance system, also known as a high-level controller in some literature, is responsible for directing an autonomous sailboat to behave as desired. Typically, the missions of autonomous sailboats (and other USVs) can be classified into two types: sailing and station-keeping. Sailing involves maneuvering an autonomous sailboat from one position to another, while station-keeping is the task of keeping an autonomous sailboat in a specified area, usually defined using a point with a radius.

To achieve these missions, a guidance system generates reference signals based on the given tasks and data received from a navigation system. Two common reference signals are the desired course angle χ_d and the desired speed U_d (Sun et al., 2018). In some cases, the desired heading angle ψ_d and the desired surge speed u_d (further explained in Section 4.2) may be used instead of these reference signals. Fig. 11 illustrates the guidance system diagram of an autonomous sailboat.

To go from one point to another, a path or trajectory that connects the two points must be found. A path is simply a set of points connecting the start and endpoint without specifying the speed of travel along the path. In contrast, a trajectory is a path that includes a schedule, thereby implicitly specifying speed as a function of time. Since a reference signal is a function of time, it can also be considered as a trajectory. However, for the sake of clarity in surrounding contexts, the terms "path" and "trajectory" can be used interchangeably.

To generate proper values for a control system, a guidance system must perform path planning. Two distinct types of path planning are available: global path planning and local path planning (Liu et al., 2016). The following two subsections will clarify these concepts.

4.1. Global path planning

Global path planning is responsible for determining the optimal path or trajectory for an autonomous sailboat to travel from a starting waypoint to a final waypoint based on a given mission. The literature provides several methods to achieve this goal in sailing, such as A* (Erckens et al., 2010), Dijkstra (Zyczkowski and Szlapczynski, 2023), PRM-Dijkstra (Saoud et al., 2015b), Improved Ant Colony Optimization (IACO) (Nian et al., 2021), improved adaptive ant colony optimization and rolling window method (RIACO) (Shen et al., 2023), Improved Gray Wolf Optimization (IGWO) (Wang et al., 2019), Improved Particle Swarm Optimization (IPSO) (Gong et al., 2020), optimal Rapidly Exploring Random Tree (RRT*) (Ulysse et al., 2019), Sailing Boat Path (SBP) based on dynamic programming (Sidoti et al.,

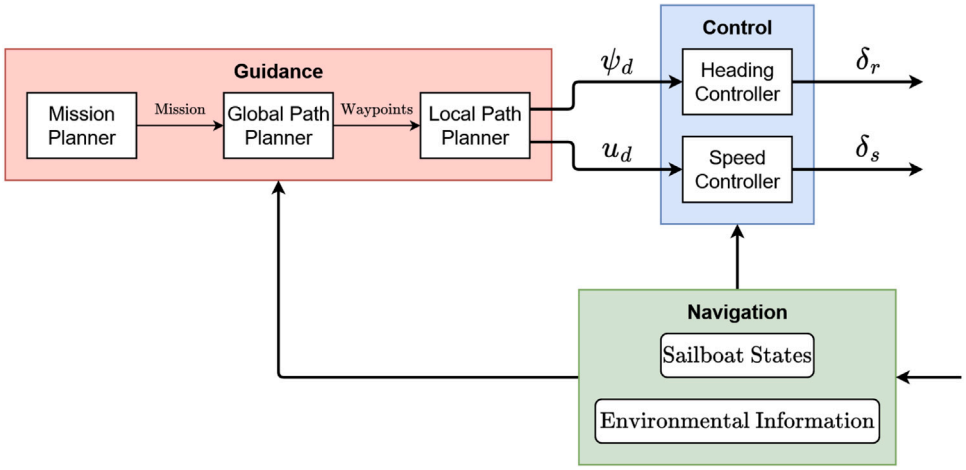


Fig. 11. Autonomous sailboat guidance system diagram.

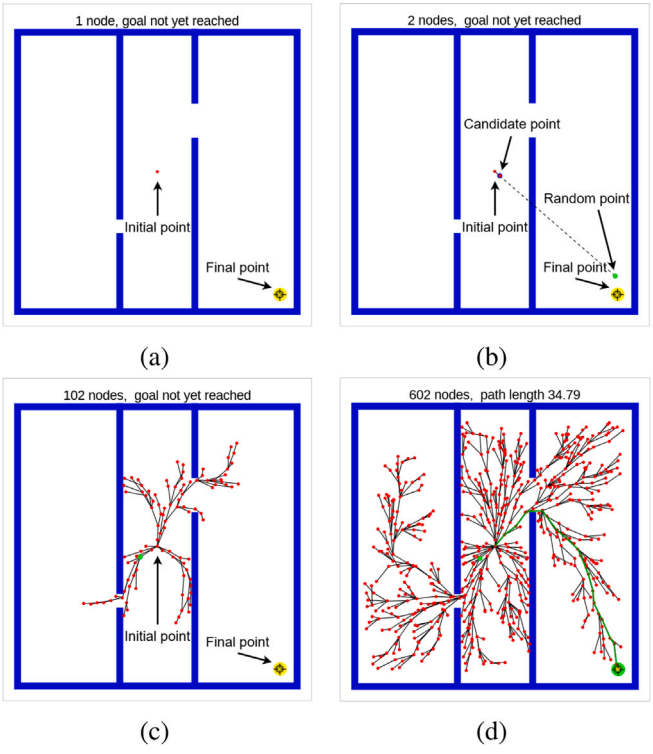


Fig. 12. Path generated by RRT* algorithm: (a) initializing; (b) checking the candidate point; (c) spreading; (d) reaching the final point. (These figures were generated from Huang (2021).)

2023), Q-Learning (Silva Junior et al., 2020), and Gaussian process based Q-learning (GP-QL) (Liu et al., 2022b).

For example, the RRT* algorithm begins with a graph containing an initial and a final point. A random point is generated; then, a candidate point is created in the same direction as the random point and within a defined radius of the graph. If the edge between the candidate point and the nearest point in the graph has no obstacle, the candidate point is added to the graph. Otherwise, the candidate point is discarded. This process is repeated to expand the graph until a feasible path from the initial point to the final point is established (Karaman and Frazzoli, 2011), as illustrated in Fig. 12.

4.2. Local path planning

Unlike global path planning, local path planning transforms the path given by global path planning into several waypoints and considers only the ongoing path or trajectory while handling dynamic changes in the environment such as obstacles or wind direction (Marin-Plaza et al., 2018). In sailing, this includes short-term path planning for tacking and jibing. As a result, local path planning generates the proper desired heading and speed, which will be sent to a control system afterward. Methods used to achieve this goal include: Line-of-Sight (LOS) (Yang et al., 2021; Lee Boyce and Hugh Elkaim, 2007; Deng et al., 2020a; Plumet et al., 2013) and Potential Field (PF) (Ulysse et al., 2019; Plumet et al., 2013, 2015; Pêtrès et al., 2011, 2012; Liu et al., 2016).

For example, the LOS method calculates the desired course angle χ_d given by Eq. (33) based on the line between the current waypoint P_k and the next waypoint P_{k+1}, the lateral distance e, and the predetermined lookahead distance Δ as shown in Fig. 13.

$$\chi_d = \chi_p + \chi_r \tag{33}$$

$$\chi_r = \arctan\left(\frac{e}{\Delta}\right) \tag{34}$$

Generally, the surge speed u is much higher than the sway speed v; consequently, the sideslip angle can be neglected ($\beta = 0$) and the course angle is equal to the heading angle ($\chi = \psi$) (Yang et al., 2021; Fossen, 2011).

In sailing, there is an algorithm called the line following control that contains a guidance part, which computes the desired heading according to the line to be followed and includes consideration of tacking (Jaulin and Le Bars, 2013b). The control part and other details of the line following control will be described in Section 5.1. Inputs for the guidance part are: the line to be followed, defined using the start point $\boldsymbol{a} = [a_x, a_y]^\top$ and the final point $\boldsymbol{b} = [b_x, b_y]^\top$; sensor data consisting of the sailboat position $\boldsymbol{m} = [x, y]^\top$, the sailboat heading ψ, and the true wind angle α_{tw}; and tuning parameters consisting of the close hauled angle ζ, the incidence angle ψ^∞, and the cutoff distance c. The algorithm computes the lateral distance $e = \det\left(\frac{\boldsymbol{b}-\boldsymbol{a}}{\|\boldsymbol{b}-\boldsymbol{a}\|}, \boldsymbol{m}-\boldsymbol{a}\right)$ and the line angle $\psi_{ab} = \arctan2(b_y - a_y, b_x - a_x)$. The candidate heading angle ψ^* is given by:

$$\psi^* = \psi_{ab} - \frac{2\psi^\infty}{\pi}\arctan\left(\frac{e}{c}\right). \tag{35}$$

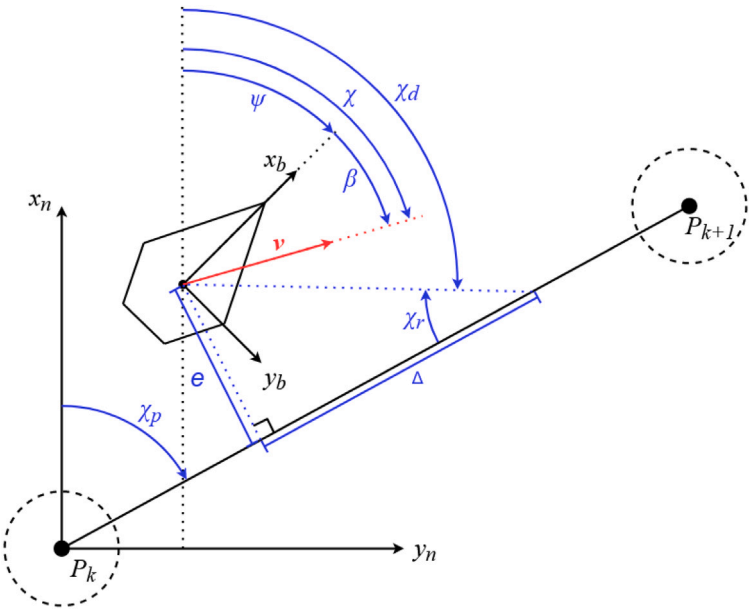


Fig. 13. LOS method vectors.

Next, the desired heading angle is decided depending on the maneuvering mode (nominal or tack) by

$$\psi_d = \begin{cases} \pi + \alpha_{tw} - d\zeta & \text{if } \cos(\alpha_{tw} - \psi^*) + \cos(\zeta) < 0 \\ & \text{or } \left(|e| < c \text{ and } \left(\cos(\alpha_{tw} - \psi_{ab}) + \cos(\zeta) < 0\right)\right) \\ \psi^* & \text{otherwise} \end{cases} \tag{36}$$

where $d = \text{sgn}(e)$ is the state depending on the port or starboard tack when the sailboat exceeds the cutoff distance (Melin et al., 2016).

As described above, the guidance system generates the desired heading ψ_d and speed u_d, which are sent to the heading and speed controllers, as shown Fig. 11. These controllers have several control strategies that are introduced in the following section.

5. Motion control strategies

A control system, called a low-level controller in some of the literature, is responsible for controlling the behavior of an autonomous sailboat provided by the guidance system. For both sailing and station-keeping, the two primary control inputs are the rudder angle δ_r and the sail angle δ_s. The rudder angle δ_r is used to control the course angle χ or heading angle ψ, while the sail angle δ_s is used to control the sailboat speed U or surge speed u. The control system diagram for an autonomous sailboat is shown in Fig. 14. With the exception of catamarans, the roll angle ϕ must also be taken into consideration to maintain the sailboat's heeling and prevent capsizing, as it is a sailing constraint. This problem can be addressed by adjusting the sail angle of attack since the sailboat's heeling is primarily affected by the sail. Notably, the sailboat's dynamics represent a Multiple-Input Multiple-Output (MIMO) system. However, the rudder angle and the sail angle are usually simplified by being controlled independently from each other. While the sailboat's heading is closed-loop controlled by its rudder angle, the sailboat's speed is typically open-loop controlled by the sail angle.

The following subsections will cover the heading control strategies based on the rudder angle δ_r and the speed control strategies based on the sail angle δ_s (more detail is provided in Section 5.1 and Section 5.2, respectively). In addition, a summary of recent autonomous sailboat control strategies will be presented in Section 5.3.

5.1. Heading control strategies

The sailboat's linear velocity is composed of two components, namely u and v, representing the speed in the surge and sway directions, respectively. However, the surge speed u dominates the linear velocity primarily. Thus, the rudder angle δ_r is used to control the heading direction, which, in turn, controls the speed direction. The heading control strategies are classified into three types based on their objectives: (1) heading error minimization, (2) objective optimization, and (3) AI-based control. The heading control strategies are illustrated in Fig. 15.

5.1.1. Heading error minimization

The most basic purpose of heading control is to minimize the heading error between the current sailboat heading and the desired heading generated by a guidance system.

- **Mechanism Control: Self-Steering Mechanism (SSM)** A self-steering mechanism is a control that provides a hardware mechanism to steer a sailboat. A self-steering mechanism is implemented to actuate the rudder by the wind instead using electrical energy, which provides the advantage of saving energy. The rudder is connected to a wind vane through a linkage that adjusts the rudder if the wind direction changes (An et al., 2021). However, there is a limitation in that the heading direction can only be adapted as a constant apparent wind angle for sailing (Ulysse et al., 2019). An example of a self-steering mechanism is shown in Fig. 16.
- **Simple Control: PID Controller** A Proportional-Integral-Derivative (PID) controller is a simple method widely used for heading error control, as shown in the block diagram in Fig. 17. This method uses the desired heading angle ψ_d sent from a guidance system and the current heading angle ψ to calculate the error e, which is given by:

$$e = \psi_d - \psi. \tag{37}$$

Then, the error is used to calculate the rudder angle δ_r, which is given by:

$$\delta_r = K_p e + K_i \int_0^t e dt + K_d \frac{de}{dt} \tag{38}$$

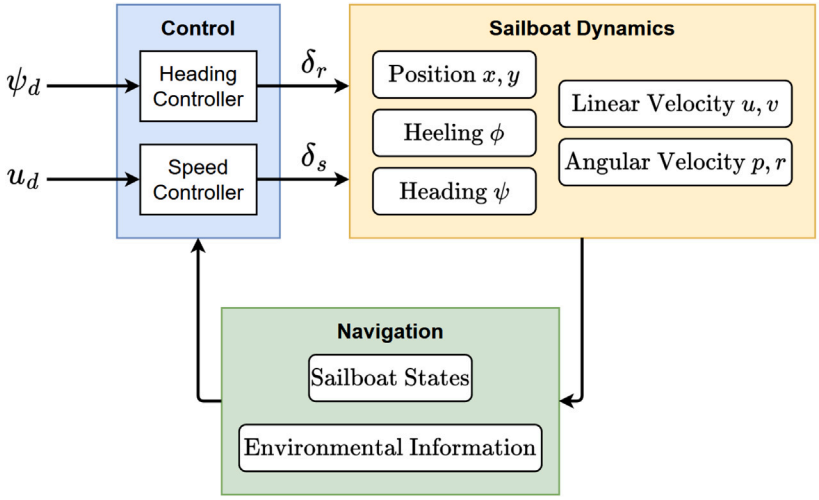


Fig. 14. Autonomous sailboat control system diagram.

where the proportional gain K_p, the integral gain K_i, and the derivative gain K_d are the parameters to be tuned. As a result, the system gives the current output (Borase et al., 2021).

Generally, a PID controller is used with constant gains for the entire sailing course. However, it can be combined with other methods, such as a Gain-Scheduling (GS) (dos Santos and Goncalves, 2019) or a fuzzy logic system (Liu et al., 2022a) to choose the gain values dynamically depending on certain conditions. Furthermore, controller performance can be tuned for optimal gains using methods, such as Ziegler–Nichols (Z–N), Particle Swarm Optimization (PSO), and Beetle Antennae Search (BAS) (Zhou et al., 2020). The Aeolus project (Tranzatto et al., 2015; Tranzatto, 2015; Wirz et al., 2016) designed a proportional controller that used a constant gain and a nonlinear (NL) proportional controller utilizing nonlinear gain to control the sailboat heading.

- **Simple Control: Line Following Control (LFC)** The other simple control strategy for sailing is called the line following control (Jaulin and Le Bars, 2013b), which consists of guidance (as described in Section 4.2), heading control, and speed control parts to allow a sailboat to follow a defined line. The speed control part takes the concept of the linear sail angle (described later in Section 5.2.2). The line following control is a simple control strategy recommended for a beginner to control a sailboat since it is uncomplicated to follow and its algorithm can be understood without requiring knowledge of sailboat dynamics. The stability analysis of a line following controller was described in Jaulin and Le Bars (2013a).

  In this algorithm, after the desired heading angle is computed, the rudder angle is determined proportionally to the sine function of the heading error with a maximum limit and given by:

$$\delta_r = \begin{cases} -\delta_r^{max} \sin(\psi - \psi_d) & \text{if } \cos(\psi - \psi_d) \geq 0 \\ -\delta_r^{max} \operatorname{sgn}\big(\sin(\psi - \psi_d)\big) & \text{otherwise} \end{cases} \tag{39}$$

  where δ_r^{max} is the maximum rudder angle. Notably, the original line following control is defined in the East-North-UP (ENU) coordinate system. Hence, left negative signs appear in Eq. (39) when the rudder angle is instead defined in the NED coordinate system.

  The Åland Sailing Robots (ÅSR) project (Melin et al., 2016) found that the line following controller produced the worst time and performance (indicated by $\Sigma|\psi - \psi_d|$ and $\Sigma|e|$) compared to the P and PD controllers in the study. Nevertheless, the control signal activity (indicated by $\Sigma|\delta_r|$) was lowest for the line following controller, followed by the P and PD controllers.

- **Nonlinear Control: Feedback Linearization (FL)** The feedback linearization method transforms a nonlinear system into a controllable linearized system using nonlinear feedback and a nonlinear coordinate transformation in the inner loop. Then, a linear feedback control method is applied in the outer loop (Isidori, 1995; Khalil, 2002). The block diagram of the heading control with the feedback linearization controller is shown in Fig. 18.

  In the literature, various heading controller designs have been proposed using feedback linearization. A study in Wille et al. (2016a) combined the feedback linearization in the inner loop and the PD controller in the outer loop, using the heading error and the turning rate. Another study in Le Gallic et al. (2018) proposed feedback linearization based on a vector field approach in the inner loop and then the pole placement (PP) method for the linearized system in the outer loop. Due to some singularities of the feedback linearization, the work in Herrero et al. (2010) combined the feedback linearization with a pre-compensator that mapped the set-point of speed and heading required by the user to an acceptable set-point of sail angle and heading by the controller. In Abrougui et al. (2019b), feedback linearization was used for both heading and speed simultaneously in the inner loop, and the PDD^2 controller was designed for the outer loop.
- **Nonlinear Control: Backstepping Control (BC)** Backstepping control is a control strategy that stabilizes a new state defined by the states of a nonlinear system using a virtual control input chosen recursively under Lyapunov stability (Khalil, 2002). This method provides guaranteed stability and can handle uncertainties, resulting in high-quality control performance even if some parameters vary. However, it is plagued by the issue of "explosion of terms" or "explosion of complexity" caused by the virtual controller's recurrent derivatives (Zhang et al., 2019b; Li et al., 2022). To circumvent this problem, recent studies (Zhang et al., 2019a, 2022; Yingjie et al., 2019) have applied Dynamic Surface Control (DSC), which simplifies the backstepping control method by introducing first-order filters. The addition of these filters enables using the virtual control design without differentiation (Swaroop et al., 2000; Wang and Huang, 2005). Thus, we consider DSC as a backstepping control method.

  A nonlinear heading controller for an autonomous sailboat was proposed in Xiao and Jouffroy (2014) using the backstepping control method. Additionally, the same study proposed a moving mass system for course control with some parameters that required tuning. Another study in Wille (2016) simplified the backstepping approach for course control. Due to uncertainties present in the backstepping control law, they were approximated

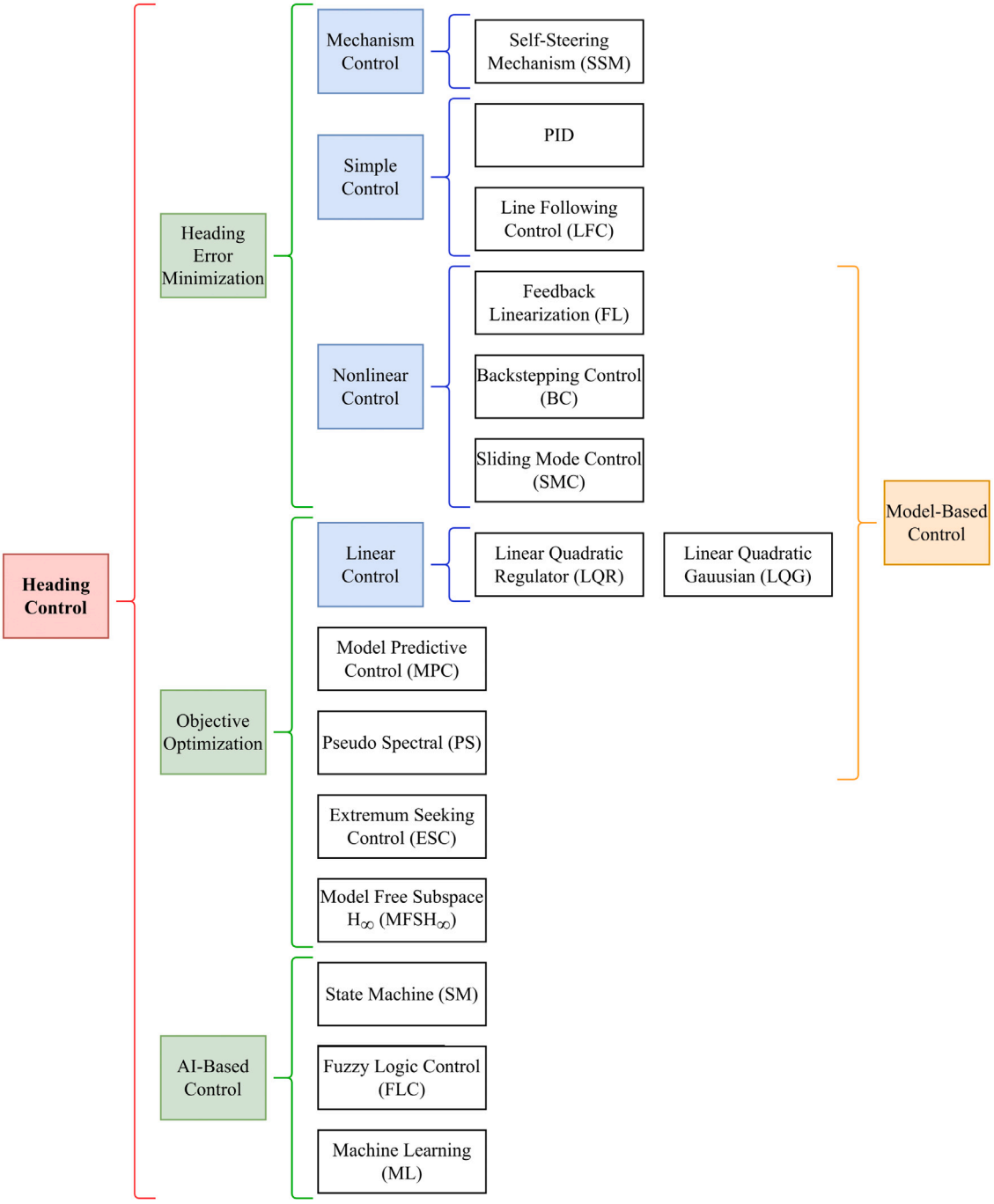


Fig. 15. Heading control strategies: Heading error minimization, Objective optimization, and AI.

using fuzzy logic systems in some studies, such as (Deng et al., 2019; Yingjie et al., 2019; Deng et al., 2020b; Zhang et al., 2022, 2019a). However, other studies, such as (Deng et al., 2020a; Zhang et al., 2023), addressed the same issue using adaptive neural methods, such as Echo State Networks (ESNs) and Radial Basis Function Neural Networks (RBF-NNs).

- **Nonlinear Control: Sliding Mode Control (SMC)** A sliding mode control is a nonlinear control method used to design a sliding surface $S = 0$ to satisfy Lyapunov stability. There are two phases: (1) a reaching phase, which is the phase of moving the system state to the sliding surface, and (2) a sliding phase, which is the phase of sliding the system state along $S = 0$ to desired states (Khalil, 2002). Although the sliding mode control has the advantage of robustness, the chattering effect produced by the discontinuity in actuating commands is a drawback to be improved (Erbatur et al., 1999).

  A heading controller using sliding mode control combined with feedback linearization corresponding to the pole placement method was designed in Abrougui et al. (2019a), Abrougui and Nejim (2018b). Due to the chattering effect caused by the signum function, they replaced it with the arc-tangent function for better results.

5.1.2. *Objective optimization*

Despite the heading error minimization purpose, other objectives such as optimal tacking, optimal energy consumption, and heeling constraint are usually considered to control an autonomous sailboat.

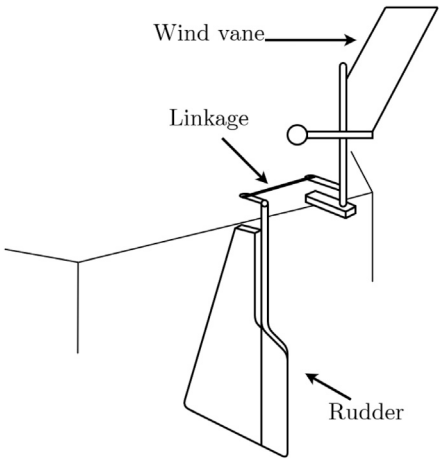


Fig. 16. Self-steering mechanism. *Source:* Adapted from Offshore Sailor: Windvane self steering (2017).

Control strategies in this category deal with an objective function defined for the required objective.

- **Linear Control: Linear Quadratic Regulator (LQR)** The LQR control provides optimal control using an appropriate feedback gain according to the defined weighting matrices of states and inputs. Even so, the LQR requires a controllable and observable linear or linearized system to be applied (Fossen, 2011). To improve the controller performance, the user needs to tune weighting matrices.
  The Aeolus project (Tranzatto et al., 2015; Tranzatto, 2015; Wirz et al., 2016) used an LQR method to design an unconstrained optimal tack controller that performed faster tacking compared to their nonlinear proportional controller, with the yaw rate and the heading error being zero at the final time without overshoot. Notably, the controller could be applied due to the linear dynamic model of yaw motion identified in the study.
- **Linear Control: Linear Quadratic Gaussian (LQG)** The LQG extends an LQR method by including Gaussian noise in the system measurement. Thereby, the objective optimization is derived through the optimal expected value.
  The Atlantis project Elkaim (2006) designed an LQG controller with the Observer Kalman IDentification (OKID) model based on observed noise statistics to perform sailing with the minimized weighted sum of the cross-track error and the rudder rate.
- **Model Predictive Control (MPC)** The MPC extends the LQR method by re-initializing the initial state as measured and re-optimizing the objective function in real-time. Consequently, an MPC controller requires high-performance hardware to operate successfully. The block diagram of heading control with an MPC controller is shown in Fig. 19.
  In addition to the LQR, the thesis of the Aeolus project (Tranzatto, 2015) used MPC to design a constrained optimal tacking controller that included some constraints and a terminal term in the objective function to ensure closed-loop stability. Energy-efficient sailing using an MPC method was presented in Tréhin et al. (2020). Tacking, waypoint following, and station-keeping using MPC methods were presented in Vautier (2021). The study by Dong et al. (2021) took the Gaussian model into account with an MPC, which was called the Gaussian Process Model Predictive Control (GPMPC), to improve the tracking accuracy where there was noise, with a performance comparison of a Back Propagation Neural Network Model Predictive Control (BPMPC) and a static PID controller. The study by Liu et al. (2023) proposed a collaborative MPC that controlled the rudder and sail simultaneously for path tracking, while maintaining the roll angle.
- **Pseudo-Spectral (PS)** The PS optimal control method is a collocation method used to determine an optimal control input through a nonlinear programming problem. The main advantage of the PS is that it acquires the global optimal solution due to an approximation by a high-order polynomial (Kelly, 2017).
  The OceanVoy project (Sun et al., 2020) managed hybrid energy optimization for a rudder, sail, and other components using the PS optimal control method by considering two objective functions: the maximum high wind speed and torque optimization.
- **Extremum Seeking Control (ESC)** The ESC method adjusts the control input through the output gradient to achieve the optimal objective without requiring knowledge of sailboat dynamics. To date, (Treichel and Jouffroy, 2010) is the only study we have found that used ESC for heading control. They proposed a multi-parameter extremum search heading control coupling with the sail control that finds the best speed direction with optimal sail adjustment through a perturbation-based ESC.
  Nevertheless, ESC is applied for speed control more often than heading control. Further details of the ESC method are provided in Section 5.2.2.
- **Model Free Subspace $\mathcal{H}_\infty$ (MFS$\mathcal{H}_\infty$)** A study of the Atlantis project (Elkaim et al., 2006) proposed a novel controller called the MFS$\mathcal{H}_\infty$ optimal controller. Model-free control design techniques can control a dynamic system without the explicit model, by including two steps of design: plant identification and control design (Woodley et al., 2001). In this study, an open-loop experiment was set up to obtain data that was used to calculate a high-order predictor. Chosen parameters were used to formulate a finite horizon cost function with weighting functions. Lastly, the controller generated the optimal control input that satisfied the formulated finite horizon cost function in a receding horizon.

5.1.3. AI-based control

Aside from the mechanism control and control theories described above, Artificial Intelligence (AI)-based control strategies have been applied for the heading control of an autonomous sailboat. AI can simulate human intelligence (Chowdhury and Sadek, 2012). Therefore, AI-based control strategies have the advantage of not requiring a sailboat dynamic model to control an autonomous sailboat.

- **State Machine (SM)** State machine divides the states to be controlled into a finite number of states and defines the rules of state transition. Each state has its appropriate control law for the current state. Therefore, the control performance depends on defined rules of state transition and control laws.
  The Iboat project (Briere, 2008; Briere et al., 2009) separated maneuver states into three: normal sailing, tacking, and jibing. A simple proportional controller manages the heading control in the normal sailing state. In tacking or jibing states, the rudder is set to a constant absolute angle and switches its direction according to the port or starboard tacking or jibing. The study by Sun et al. (2018) separated the tacking maneuver into five states, as shown in Fig. 20: initial state, right turning, left turning, no wind, and accelerating. Both heading and sail angles are set manually in the initial state. The sailboat state is changed to the left or right turning state if the sailboat is on the left or right side of the pool with an appropriate heading angle, while the rudder angle is set to its maximum absolute angle in the left or right direction. The no wind state occurs if the sailboat is not in the wind zone, and the rudder angle remains the same. In neither of the above states, the sailboat is in the accelerating state, and the rudder angle is set to zero.

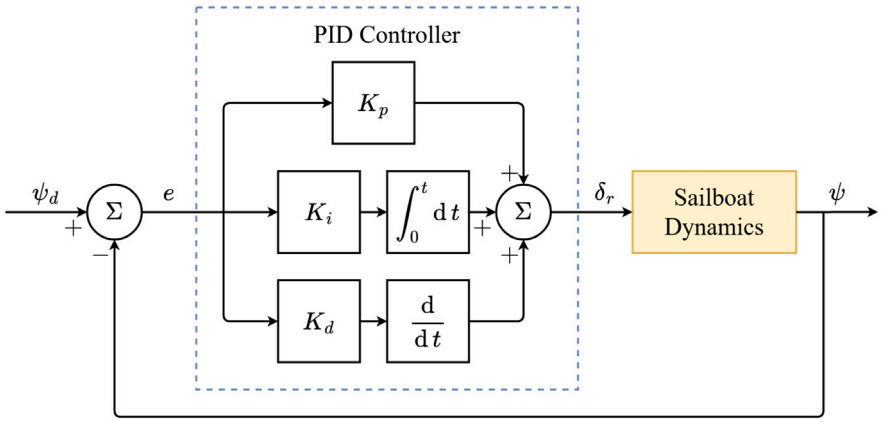


Fig. 17. PID block diagram for heading control.

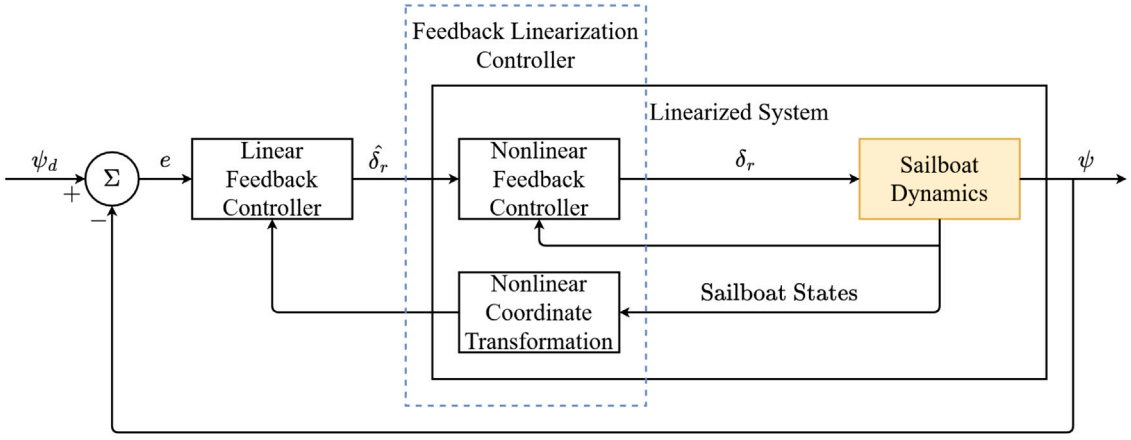


Fig. 18. Feedback linearization controller for heading control.

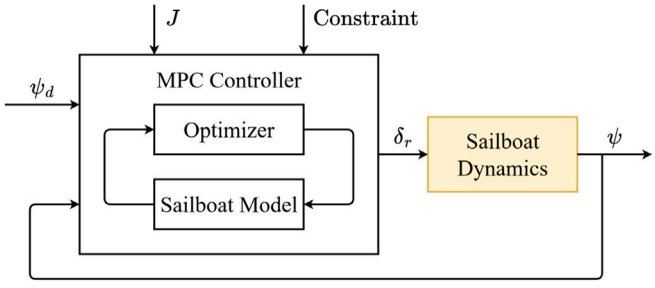


Fig. 19. MPC block diagram for heading control.

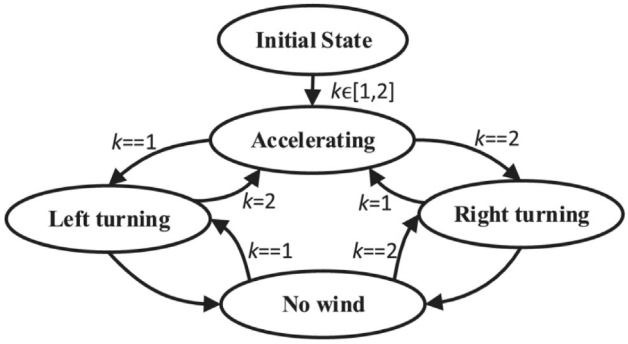


Fig. 20. Tacking maneuver state machine diagram (Sun et al., 2018).
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- **Fuzzy Logic Control (FLC)** The FLC method simulates experts' knowledge to control a system by constructing fuzzy control rules as control laws. Unlike binary logic, fuzzy logic contains a membership function with the greatest grade being one, and the least grade being zero (Takagi and Sugeno, 1985). The performance of this strategy depends on defined fuzzy rules and fuzzy sets.
  Several studies (Stelzer et al., 2007; Yeh and Bin, 1992; Liu et al., 2018; Wang et al., 2016; Gomes et al., 2017; Martins, 2013; Ramos de Miguel, 2013) have used FLC with 'if-then' rules based on sailors' experience to control the sailboat heading using the rudder angle as input. For example, if the desired direction is "left" and the turn is "neutral", then the delta rudder is "left". Fuzzy sets are used to define the terms "left", "neutral", and "right" in their respective variables. In addition to direct control, some studies in Deng et al. (2019), Yingjie et al. (2019), Deng et al. (2020b), Zhang et al. (2022) applied the fuzzy logic system to approximate uncertainties in the control law derived from another control strategy.
- **Machine Learning (ML)** ML methods, including the Neural Network (NN) method, have been proposed for use in sailing. ML methods are a subset of AI methods that can learn patterns from data to achieve the performance required by using computational algorithms (Helm et al., 2020; Jiang et al., 2020). Thus, control laws could be constructed to sail an autonomous sailboat by learning from existing data.
  The Robosail project aimed to create a semi-autonomous sailboat with an intelligent system having the ability to learn how to steer a sailboat optimally in conjunction with the human sailor onboard (van Aartrijk et al., 2002). Their control strategy focused

on AI-based methods—usually ML methods. In 2002, the Robosail project (van Aartrijk et al., 2002) attempted to train a model using Reinforcement Learning (RL), which would then be used for rudder control. In addition, for rudder control, they took into account an adaptive PID controller with Z–N open-loop tuning. In 2003, the Robosail project (Adriaans, 2003) used a neural network combined with a PID controller for rudder control. Another study of the Robosail project in the same year (van Aartrijk and Samoocha, 2003) extended sailing jargon to fuzzy sets and used generated training samples to develop decision trees in the rudder control system.

The study by Fossati (2003) provided batch learning with the back-propagation algorithm to train a neural network, which was then applied in the rudder control system. In addition, they utilized the adaptive learning rates approach for faster learning and to guarantee convergence. Another study (Astrov and Udal, 2020) used a neural network predictive controller to predict future sailboat behavior that was highly nonlinear, and produced optimal control signals for those controlling the sailboat heading. Similar to the FLC approach that can be applied for the approximation of uncertainties, a combination of control strategies between an adaptive control and a neural network (called an adaptive neural control) that utilizes the NNs' approximation ability and stabilizes complex nonlinear systems through an adaptive control technique (Liu et al., 2011). In general, the RBF-NNs are effective for uncertainties approximation in nonlinear control, as discussed in Zhang et al. (2019a, 2020, 2023). ESNs were utilized for the same purpose in Deng et al. (2020a) as the ESN contains the capacity to learn fast-jumping and slow-varying components.

5.2. Speed control strategies

The generation of sailboat speed is primarily dependent on the wind's action on the sail, which can be manipulated by adjusting the sail angle to modify the angle of attack. As controlling a sail directly affects a sailboat's speed, some literature has referred to sail control instead of speed control. In this paper, we present a categorization of sailboat speed control strategies based on their computational approach, dividing them into three categories: (1) offline computation, (2) online computation, and (3) AI-based control, as shown in Fig. 21.

5.2.1. Offline computation

Offline computation strategies for sailboat speed control do not require real-time computation of sail adjustments. Instead, these strategies precompute or predefine a control law that can be applied to the sail immediately. Due to this advantage, offline computation strategies typically require less processing time in the onboard controller.

- **Mechanism Control: Fixed Sail Angle (FSA)** Setting a fixed sail angle for the entire sailing course is the simplest way to generate propulsion for a sailboat. However, not every wind direction can be utilized to achieve the maximum surge speed when using a fixed sail angle. For optimal performance, the fixed sail angle is recommended to be set between 30° and 50° degrees, allowing the sailboat to sail closer to the wind (dos Santos and Garcia Gonçalves, 2020). While this strategy cannot maintain optimal sailboat speed if the wind direction changes, it has the advantage of low energy consumption since sail actuation is not required.
- **Mechanism Control: Self-Trimming Mechanism (STM)** In the STM strategy, a sailboat's conventional sail is replaced with a self-trimming wingsail that can freely rotate with the apparent wind. The wingsail is controlled by a tail mounted behind it, which can adjust the angle of attack to generate the torque around the wingsail, resulting in generated force from the wingsail's angle of attack (Ulysse et al., 2019), as shown in Fig. 22. If the tail is set at a constant deflection, the wingsail will be adjusted at a constant angle of attack, allowing the sailboat to sail with a constant angle of attack even in a slight change of wind direction. However, there is a delay in the sail angle response due to actuation by the wind and the inertia of the wingsail. In addition, the self-trimming wingsail has poor performance in downwind sailing, since the tail cannot adjust to the correct sail angle (Walker, 1985). Nonetheless, a self-trimming wingsail can be combined with another control strategy to change the angle of attack in real-time. The tail does the actuation, resulting in reduced energy consumption and a substantially lower control frequency (An et al., 2021).
- **Lookup Table (LT)** A simple strategy for sail adjustment is to use predetermined data from a lookup table that provides the appropriate sail angle for the current apparent wind. The lookup table can be constructed using calculated data, data obtained from experiments, or a polar diagram. However, the drawback of this strategy is its high space complexity, as it requires a large amount of predetermined data. To overcome this, an interpolation method is usually applied in real-time when the current state falls between discrete states in the lookup table.

  In the literature, the FASt project (Cruz and Alves, 2014) and the Avalon project (Erckens et al., 2010) have controlled the angle of attack based on a lookup table obtained from experiments. The SailBot project (Williamsz et al., 2014) assigned four discrete states of the sail based on the apparent wind direction. The study in Wille et al. (2016b) computed optimal sail angles based on the desired moment and apparent wind direction offline and stored them in a lookup table for later use.

5.2.2. Online computation

Online computation strategies involve real-time computation of proper sail adjustment to maximize the sailboat surge speed or to track the desired speed. To achieve these objectives, the sailboat states and environmental information are required in the computation. Therefore, typically, these strategies require more processing time in an onboard controller compared to offline computation strategies.

In speed control, the FL approach shares the same purpose as in heading control–to transform a nonlinear model into a controllable linear model. Therefore, the description of FL is not included in this section.

- **Linear Sail Angle (LSA)** The LSA strategy proposes a simple function for adjusting the sail angle based on the point of sail, as illustrated in Fig. 3. Essentially, the sail is opened less as the sailboat heads closer to the incoming wind. While the LSA function is often linear, it can be adjusted to fit the sailboat's characteristics and the sailing conditions. The minimum and maximum sail angles are typically set to 0° and 90°, respectively, and the sail angle is interpolated between discrete points based on the apparent wind direction. A simple function of LSA, based on the apparent wind angle, can be expressed by:

$$\delta_s = (\delta_s^{max} - \delta_s^{min})\frac{\alpha_{aw}}{\pi} + \delta_s^{min} \tag{40}$$

  where δ_s^{max} is the maximum sail angle and δ_s^{min} is the minimum sail angle (Santos et al., 2016; dos Santos and Garcia Gonçalves, 2020). Generally, this strategy is used in sailboats because it does not require any searching computation for the optimal sail angle and the result is acceptable for sailboat propulsion. Some literature has formulated a function of the linear sail angle in a different way; however, the concept of the point of sail has been kept.

  The line following controller proposed in Jaulin and Le Bars (2013b), as applied in the VAIMOS (Le Bars and Jaulin, 2013), Marius (Naveau et al., 2013), and ÅSR (Melin et al., 2016) projects, adjusted the maximum sail angle according to a cardioid

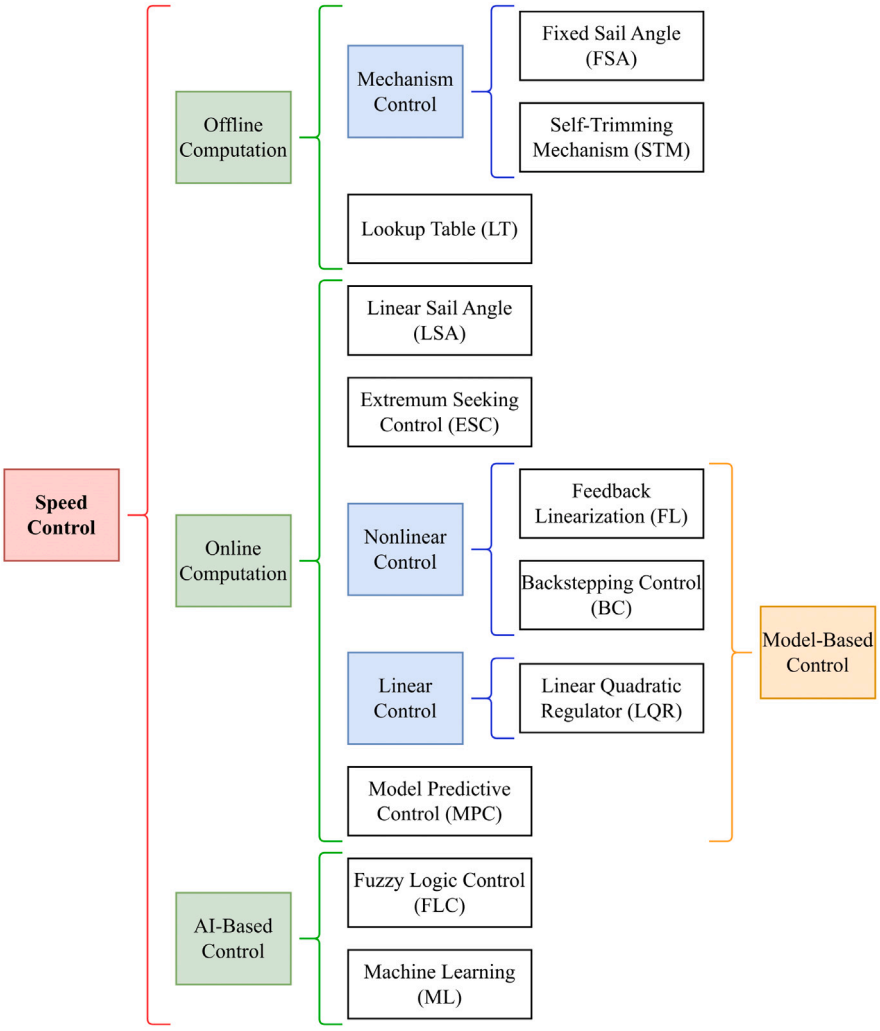


Fig. 21. Speed control strategies: Offline computation, Online computation, and AI.

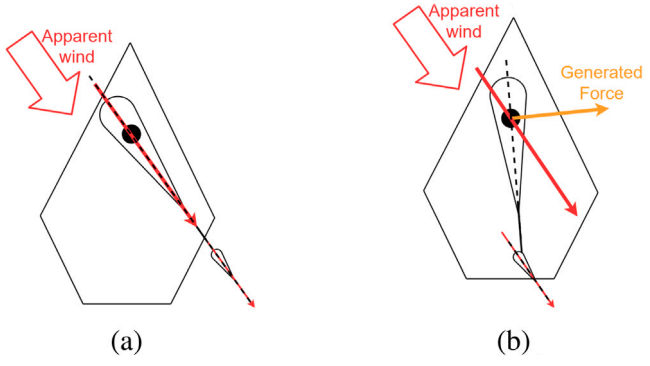


Fig. 22. Self-trimming wingsail: (a) while tail is aligned with wing; and (b) while tail is not aligned with wing.

curve (Vautier, 2021), which is a curve similar to the polar diagram. This adjustment is required because the line following controller does not directly control the sail angle directly but rather the mainsheet, which is usually controlled through servo winches. The adjustment is given by:

$$\delta_s^{max} = \frac{\pi}{2}\left(\frac{\cos(\alpha_{tw} - \psi_d) + 1}{2}\right). \tag{41}$$

Other studies on LSA, such as those in Abrougui et al. (2019b,a), Abrougui and Nejim (2018b), formulated the reference sail angle based on the polar diagram. In the SAILAMRS project (Yang et al., 2021), the heel angle was additionally taken into account for sail adjustment to ensure safety.

- **Extremum Seeking Control** The ESC is a strategy that optimizes an objective function by observing the output of the system resulting from the input, without requiring prior knowledge of the system's behavior. However, ESC requires a convergence time to reach the extremum objective (Nešić, 2009). In sailboat speed control, this control strategy is applied to maximize the actual sailboat speed without the sailboat dynamic model. Two types of ESC are typically used: perturbation-based ESC and non-perturbation-based ESC. Both perturbation-based and non-perturbation-based ESCs are applied in real-time operation. In a perturbation-based ESC, the control input optimizes an objective function that depends on the system output, while a perturbation ($a\sin(\omega t)$) is applied to an unknown system to inspect gradients. The steady-state optimization of a nonlinear dynamical system is given by:

$$\min_{\boldsymbol{u}} J(\boldsymbol{x}, \boldsymbol{u}) \quad \text{subject to } \dot{\boldsymbol{x}} = F(\boldsymbol{x}, \boldsymbol{u}) \equiv 0 \tag{42}$$

where J is the objective function, $\boldsymbol{x}$ is the system state, $\boldsymbol{u}$ is the control input, and F is a smooth function describing the dynamics (Dochain et al., 2011). In sailboat speed control, the control input is the sail angle δ_s and the output is the surge speed u.

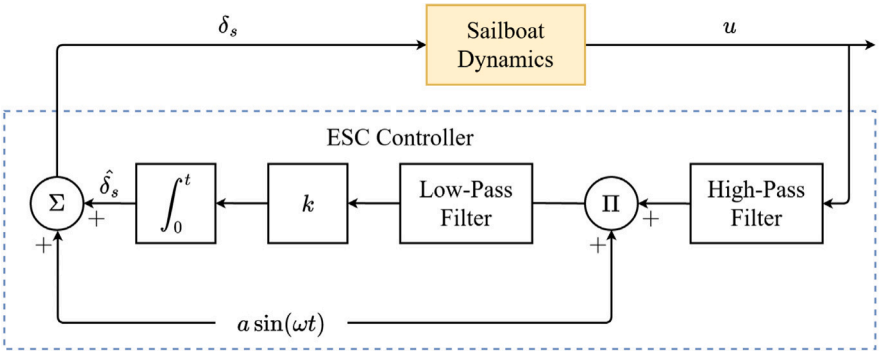


Fig. 23. Perturbation-based ESC block diagram for speed control.

The block diagram of speed control with perturbation-based ESC is shown in Fig. 23. On the other hand, non-perturbation-based ESC performs an optimization without perturbation. Instead, this method adjusts the control input according to the signs of the recent output and applied input that act as gradients.

General uses of perturbation-based ESC for sail adjustment can be found in Wirz et al. (2016), Treichel and Jouffroy (2010), Corno et al. (2016). The OceanVoy project (Sun et al., 2020) took the polar diagram as a reference sail angle for the ESC controller. Due to oscillation in the steady-state problem, dos Santos and Garcia Gonçalves (2020) proposed an improved strategy called Improved Extremum Seeking Control (IESC). This strategy combines the ESC with a PI controller and sets the sampling time to be variable for greater control, while allowing adjustment of lower and higher frequencies. The study in Shen et al. (2019) proposed an Extremum Seeking Without Steady-state Oscillation (ESWSO) to control the sailing speed, without the steady-state oscillation, by using an additional feedforward part. Furthermore, they proposed a novel speed controller in Shen et al. (2021) called Sliding Mode Extremum Seeking Control (SMESC), which performs faster convergence without steady-state oscillation. A non-perturbation ESC, which is a modified hill-climbing algorithm used for sail adjustment, was presented in Xiao et al. (2012).

- **Nonlinear Control: Backstepping Control (BC)** BC is a closed-loop control approach that has the ability to stabilize a desired state for a nonlinear system. Because of this, it is possible to derive the control law for a sail to enable the speed tracking of an autonomous sailboat by stabilizing the speed error (see details of BC in Section 5.1.1).

  In Zhang et al. (2022), the controller was designed by the DSC, which simplifies the backstepping approach, when a fuzzy logic system was used to approximate the uncertainties. In Deng et al. (2019), the fuzzy logic system was used for the same purpose; nevertheless, the control law was derived by the BC, including a sub-technique called Event-Triggered Control (ETC), to design the triggering condition, resulting in a lower actuation frequency. In a subsequent study (Deng et al., 2020b), the controller was further applied to a 4-DOF mono-hull sailboat by taking into account roll motion.
- **Linear Control: Linear Quadratic Regulator** As previously discussed in Section 5.1.2, the LQR method involves the optimization of a linear system.

  Optimization of the surge speed and desired heel angle for an autonomous sailboat using the LQR method was introduced in Wille et al. (2016b), Wille (2016). The LQR controller was applied to a linearized sailboat model obtained using the feedback linearization method in these studies. An optimal sail angle was selected from the precomputed lookup table in the case of unconstraint sailing. If the heel angle exceeded 10 degrees, the controller adjusted the sail angle to reduce the roll motion according to the designed quadratic cost function.
- **Model Predictive Control** MPC is an efficient approach to finding control inputs that optimize a given objective while accounting for constraints, as described in Section 5.1.2.

  A recent study by Liu et al. (2023) introduced a collaborative MPC that simultaneously minimizes the cost function for optimal rudder and sail angles at each sampling time. The cost function is designed as a quadratic function that weighs outputs, such as the heading and heeling angles, and control inputs, such as the rudder and sail angles. Their simulation results showed that the collaborative control approach achieved less roll angle and lateral displacement force than a separation control approach, where the MPC was only applied to heading control without considering heeling, while the sail angle was determined using an optimal sail angle strategy.

5.2.3. AI-based control

Similar to their use in heading control, AI-based methods have been applied to speed control, with the aim of simulating human intelligence to construct the control law.

- **Fuzzy Logic Control** As described in Section 5.1.3, where FLC is applied to imitate experts' knowledge within a fuzzy logic system, it is also possible to control a sailboat's speed through prior human knowledge of sailing by applying suitable sail adjustments. The studies in Stelzer et al. (2007), Júnior et al. (2013), Yeh and Bin (1992), Gomes et al. (2017), Martins (2013), Ramos de Miguel (2013) applied FLC with 'if-then' rules based on sailors' experience to control the sailboat speed. For example, if the heeling is "too high" then "ease off" the sheets, where "too high" is defined using fuzzy sets for heeling, while "ease off" is defined using fuzzy sets for sail change. By covering a range of fuzzy rules and fuzzy sets, sailboats can effectively adjust their sails and perform sailing maneuvers as desired, based on the expert knowledge and experience captured within the FLC system.
- **Machine Learning** The definition of ML was described in Section 5.1.3. The use of ML methods in direct sailboat speed control has not been widely explored in the literature.

  The Robosail project (van Aartrijk et al., 2002; Adriaans, 2003) used the k-Nearest Neighbors (kNN) algorithm to advise the sailor on the appropriate sail trim. Another RoboSail project (van Aartrijk and Samoocha, 2003) used a SailCharter module to provide advice on optimal sail trimming, although the study did not provide further details on its design. Similar to the ML in heading control, the study in Zhang et al. (2020) utilized an ML method of RBF-NNs to approximate the uncertainties in the speed control law.

5.3. Summary of recent autonomous sailboat control strategies

Our survey of recent studies on autonomous sailboats includes a presentation of the proposed control strategies for each study, with

Table 3
Recent autonomous sailboat heading control.

Heading control strategy	Ref
SSM	Sliwka et al. (2011); Friebe et al. (2017); Ulysse et al. (2019)
P	Lee Boyce and Hugh Elkaim (2007); Sauze and Neal (2006); Tranzatto et al. (2015); Wirz et al. (2016); Tranzatto (2015); Melin (2015); Melin et al. (2016)
NL-P	Tranzatto et al. (2015); Wirz et al. (2016); Tranzatto (2015)
PD	Melin (2015); Melin et al. (2016); Plumet et al. (2013); Plumet et al. (2015)
PD & FL	Wille et al. (2016a); Shen et al. (2019)
PDD^2 & FL	Abrougui et al. (2019b)
PI	Cruz and Alves (2010); dos Santos and Garcia Gonçalves (2020); Elkaim and Kelbley (2006b); Elkaim and Kelbley (2006a); Cruz and Alves (2014); Williamsz et al. (2014); Santos et al. (2016)
PI & FL	Corno et al. (2016)
PID	Santos et al. (2016); Astrov et al. (2022); Erckens et al. (2010); Pêtrès et al. (2011); Yang et al. (2021); Shen et al. (2021); Júnior et al. (2013); Sun et al. (2023); Lemaire et al. (2019)
PID & FLC	Liu et al. (2022a)
PID & ML	Adriaans (2003)
BAS-PID	Zhou et al. (2020)
GS-PI	dos Santos and Goncalves (2019)
Z-N-PID & ML	van Aartrijk et al. (2002)
LFC	Melin (2015); Melin et al. (2016); Le Bars and Jaulin (2013); Jaulin and Le Bars (2013b); Jaulin and Le Bars (2013a); Naveau et al. (2013); Clement (2013); Vautier (2021); Vautier et al. (2018); Morge et al. (2022); Ang et al. (2022)
FL	Herrero et al. (2010)
FL & PP	Le Gallic et al. (2018)
BC	Saoud et al. (2015b); Saoud et al. (2014); Wille (2016); Xiao and Jouffroy (2014); Abrougui and Nejim (2018a)
BC & FLC	Deng et al. (2019); Deng et al. (2020b); Zhang et al. (2022); Yingjie et al. (2019)
BC & ML	Deng et al. (2020a); Zhang et al. (2019a); Zhang et al. (2023); Zhang et al. (2020)
SMC & FL	Abrougui and Nejim (2018b); Abrougui et al. (2019a)
LQR	Tranzatto et al. (2015); Wirz et al. (2016); Tranzatto (2015)
LQG	Elkaim and Parkinson (2001); Elkaim (2006)
MPC	Tranzatto (2015); Tréhin et al. (2020); Vautier (2021); Liu et al. (2023); Dong et al. (2021)
PS	Sun et al. (2020)
ESC	Treichel and Jouffroy (2010)
MFS$\mathcal{H}_\infty$	Elkaim et al. (2006)
SM	Sun et al. (2018)
SM & P	Briere (2008); Briere et al. (2009)
SM & FLC	Briere et al. (2009)
FLC	Tranzatto (2015); Briere et al. (2009); Liu et al. (2018); Wang et al. (2016); Yeh and Bin (1992); Stelzer et al. (2007); Stelzer (2012); Martins (2013); Ramos de Miguel (2013); Gomes et al. (2017)
ML	Fossati (2003); Astrov and Udal (2020)
ML & FLC	van Aartrijk and Samoocha (2003)

Table 4
Recent autonomous sailboat speed control.

Speed control strategy	Ref
FSA	Sliwka et al. (2011); Astrov et al. (2022); Astrov and Udal (2020)
STM	Friebe et al. (2017); Ulysse et al. (2019); Lee Boyce and Hugh Elkaim (2007); Elkaim and Kelbley (2006b); Elkaim and Kelbley (2006a); Elkaim and Parkinson (2001); Elkaim (2006); Elkaim et al. (2006)
LT	Sauze and Neal (2006); Cruz and Alves (2014); Williamsz et al. (2014); Erckens et al. (2010); Pêtrès et al. (2011); Dong et al. (2021); Saoud et al. (2014); Herrero et al. (2010); Deng et al. (2020a); Zhang et al. (2019a); Sun et al. (2018); Liu et al. (2018); Sun et al. (2023); Lemaire et al. (2019); Zhang et al. (2023)
LT & FL & LQR	Wille et al. (2016b); Wille (2016)
LSA	Tranzatto et al. (2015); Tranzatto (2015); Melin (2015); Melin et al. (2016); Briere (2008); Briere et al. (2009); Plumet et al. (2013); Plumet et al. (2015); Abrougui et al. (2019b); Santos et al. (2016); Yang et al. (2021); dos Santos and Goncalves (2019); Le Bars and Jaulin (2013); Jaulin and Le Bars (2013b); Jaulin and Le Bars (2013a); Naveau et al. (2013); Clement (2013); Vautier (2021); Abrougui and Nejim (2018b); Abrougui et al. (2019a); Saoud et al. (2015b); Le Gallic et al. (2018); Xiao and Jouffroy (2014); Abrougui and Nejim (2018a); Wang et al. (2016); Vautier et al. (2018); Morge et al. (2022); Ang et al. (2022)
LSA & ESC	Wirz et al. (2016)
ESC	Xiao et al. (2012); Shen et al. (2019); dos Santos and Garcia Gonçalves (2020); Shen et al. (2021); Sun et al. (2020); Treichel and Jouffroy (2010); Liu et al. (2022a); Corno et al. (2016)
ESC & FLC	Yingjie et al. (2019)
BC & FLC	Deng et al. (2019); Deng et al. (2020b); Zhang et al. (2022)
BC & ML	Zhang et al. (2020)
MPC	Liu et al. (2023)
FLC	Briere et al. (2009); Santos et al. (2016); Júnior et al. (2013); Yeh and Bin (1992); Stelzer et al. (2007); Stelzer (2012); Martins (2013); Ramos de Miguel (2013); Gomes et al. (2017)
ML	Adriaans (2003); van Aartrijk et al. (2002); van Aartrijk and Samoocha (2003)

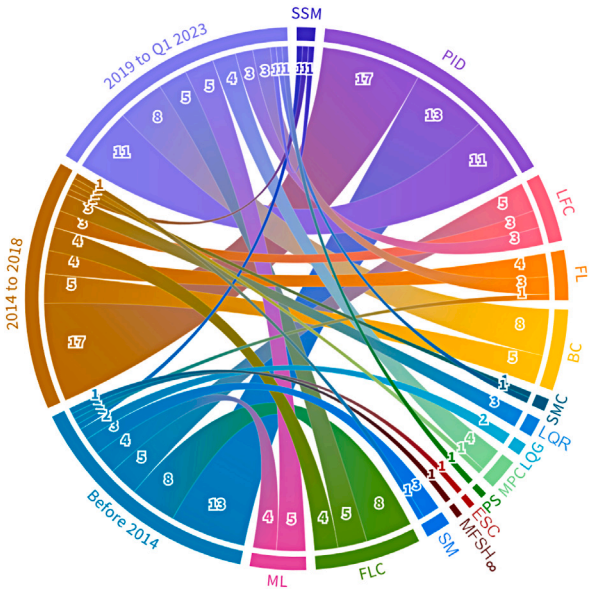


Fig. 24. Recent usage of heading control strategies.

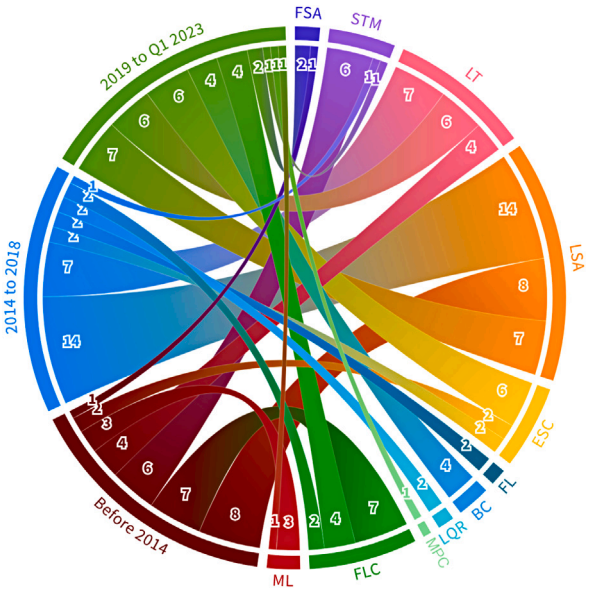


Fig. 25. Recent usage of speed control strategies.

heading control strategies listed in Table 3 and speed control strategies listed in Table 4. To assess the evolution of control strategies over time, we categorized the studies into three periods: (1) before 2014, (2) from 2014 to 2018, and (3) from 2019 to the first quarter of 2023. Figs. 24 and 25 show the frequencies of heading and speed control strategy usage, respectively. Notably, these figures may not be exhaustive regarding autonomous sailboat studies. Finally, we provide a summary of heading and speed control strategies in Tables 5 and 6, respectively, while omitting duplicate strategies to avoid redundancy.

Before 2014, heading control in autonomous sailboat studies primarily utilized model-free approaches, possibly due to the complexity of sailboat dynamics. Despite PID control being the most commonly applied strategy overall, model-free control strategies, including SSM, LFC, FL, ESC, MFS$\mathcal{H}_\infty$, SM, FLC, and ML, remained the most commonly used for heading control during that period compared to model-based strategies. From 2014 to 2018, model-based control strategies had a greater role in heading control and provided guaranteed stability through applying nonlinear control approaches, leading to a decrease in the usage of model-free strategies, except for PID. In most recent five years (from 2019 to the first quarter of 2023), there have been changes in the utilization of FLC in heading control. Unlike in the past, when FLC was primarily used to directly control the heading of a sailboat, current applications of FLC focus on approximating uncertainties within the control law derived from another control strategy, whereas MPC has emerged as a new trend in heading control due to advancements in hardware computation performance, enabling efficient implementation.

As shown in Fig. 25, regarding speed control in autonomous sailboats, LSA is the most commonly applied control strategy in the period analyzed, owing to its simplicity and acceptable efficiency. Prior to 2019, open-loop control strategies, such as FSA, STM, LT, LSA, FLC, and ML that do not measure the feedback sailboat's speed, were the main speed control methods applied to autonomous sailboats. Although ESC was introduced as a sailboat speed control method in 2010 by Treichel and Jouffroy (2010), only a few studies at that time were identified in the current survey that utilized ESC for autonomous sailboats. However, since 2019, closed-loop control strategies, including ESC, BC, and MPC, have become more popular in sailboat speed control. Although in 2016, the model-based control strategy was proposed by Wille et al. (2016b), Wille (2016), the proposed method did not directly track the desired speed but instead focus on the desired heel angle. Model-based speed tracking control is a new trend in speed controller design to track the desired speed for an autonomous sailboat, which was first introduced by Deng et al. (2019), according to our survey. Similar to heading control, current applications of FLC in sailboat speed control have changed to consider the approximation of uncertainties. In conclusion, the current trend in sailboat speed control appears to be the utilization of online computational approaches and closed-loop control strategies for speed tracking.

6. Conclusion

This survey paper has provided a comprehensive overview of autonomous sailboat control, highlighting the essential components and knowledge required for effective operation. Specifically, the paper has discussed the sailboat dynamics model and guidance and control systems, as well as discussing various control strategies for heading and speed control. By providing a clear understanding of these fundamental aspects, this paper contributes to the body of knowledge on autonomous sailboat control and provides a foundation for future research and development.

Potential future research areas on sailboat controls and technology are: (1) aerodynamics and hydrodynamics, based on the optimization of sail shapes, refining hull designs, and exploring ways to reduce drag and increase efficiency; (2) materials and composites, involving the development of lightweight but strong and durable materials; (3) energy harvesting and storage through the integration of integrating renewable energy sources and the development of efficient solar panels, wind turbines, and hydrokinetic systems to generate electricity; (4) stability and control systems, including ballast systems, stability control mechanisms, optimal routing algorithms, sensor technologies, AI and ML algorithms, and advanced control systems; (5) human–machine interaction between sailors and sailboat systems, involving the exploration of user interfaces, wearable technologies, and data visualization techniques to provide sailors with real-time feedback and decision support; and (6) environmental sustainability through the investigation of eco-friendly, antifouling coatings to reduce hull resistance and biofouling, and the development of systems to minimize the environmental impact of sailboat operations.

Table 5
Summary of heading control strategies.

Heading control strategy	Advantage/Prominence	Disadvantage/Limitation
SSM	No electricity is required.	The heading is limited to a constant wind angle.
PID	No system model is required. The most-applied heading control strategy.	Deals poorly with disturbances.
LFC	No system model is required. Easy to understand and tune parameters.	Low performance.
FL	Transforms a nonlinear system to be controllable using another linear control technique.	There must be an appropriate control to transform the system.
BC	Gains in robustness. Guaranteed stability.	Lyapunov stability is required. The "explosion of complexity" problem.
SMC	Gains in robustness. Guaranteed stability.	Lyapunov stability is required. Chattering effect problem.
LQR	Uncomplicated to implement for optimization.	Limited to a controllable and observable linear system.
LQG	Takes Gaussian noise into account in optimization.	Limited to a controllable and observable linear system.
MPC	Deals with uncertainties in optimization.	High-performance hardware is required due to high computation.
PS	Obtains a global solution in optimization.	Complicated to implement.
ESC	Optimizes the objective without requiring a system model.	Takes convergence time. Steady-state oscillation.
MFSH_∞	No system model is required. Robust.	Requires experimental data.
SM	Control each state with defined control laws.	Requires experts' knowledge.
FLC	The control performance is based on defined fuzzy rules and fuzzy sets. Good uncertainties approximation.	Requires experts' knowledge.
ML	Constructs control laws from training data. Uncertainties approximation through NNs.	Requires training data.

Table 6
Summary of speed control strategies.

Speed control strategy	Advantage/Prominence	Disadvantage/Limitation
FSA	No sail actuation and electricity are required.	The sailing direction is limited. Poor performance.
SSM	No electricity is required in sailing with a constant angle of attack. Less electricity is consumed by the actuator.	A delay in sail angle response. The sailing speed cannot reach the optimal in every sailing direction.
LT	No computation or computing, just interpolation in real-time.	Requires pre-computation. The sailing speed may not reach the actual optimal in real-time.
LSA	Low computation cost. Easy to implement with acceptable performance. The most-applied speed control strategy.	The sailing speed may not reach the actual optimal in real-time.
ESC	Reaches the optimal speed without requiring the system.	Requires convergence time. Steady-state oscillation may occur.
FLC	Performance is based on defined fuzzy rules and fuzzy sets.	Requires experts' knowledge.
ML	Constructs control laws from training data.	Requires training data.

CRediT authorship contribution statement

Yodyium Tipsuwan: Conceptualization, Writing – review & editing, Supervision, Project administration. **Peerayot Sanposh:** Conceptualization, Validation, Writing – review & editing. **Nattakit Techajaroonjit:** Conceptualization, Investigation, Writing – original draft, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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